Collected Preprints — Working Draft

Computation, Causality, and Consciousness

Practical Steps Towards Getting It from Bit

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Revision: 2026-03-15

Overarching Introduction

This work grows from a single question: *what is the minimum mathematical structure needed to ground physics, knowledge, life, and mind?*

The answer developed across four parts is that the structure of a *graph-structured lambda theory* (GSLT) — a grammar of terms, equations on those terms, and rewrite rules — is already sufficient, provided it is equipped with modest additional data. Physics emerges from weights and costs on the rewrite rules. The limits of knowledge emerge from the topology of the formula logic induced by the GSLT and the communication geometry of a massive agent population. The hypercomputational tower above any Turing-complete GSLT provides a precise setting for sentience and consciousness. Life emerges from the density of replicating fixed points in the space of processes.

None of these correspondences are metaphors. Each is a mathematical construction, stated precisely enough to be wrong. Part I is presented with considerable confidence. Parts II and III are speculative but grounded. Part IV is avowedly exploratory, reaching toward the hardest questions. All four parts share a common architecture and notation, and a common conviction: that computation, properly understood, is not merely a tool for doing science and philosophy but a substrate in which science and philosophy can be expressed with full mathematical precision.

Part I Towards a Theory of Causality

Part I asks: what is a *physics*? The answer is that a physics is a GSLT equipped with two maps — a *weight map* assigning complex amplitudes to rewrite steps, and a *cost map* assigning resource consumption — indexed by a context-decorated Hennessy–Milner logic derived canonically from the GSLT itself.

The construction proceeds in stages. Any GSLT is first lifted to a *time-symmetric envelope* $\mathcal{S}^\dagger$ in which every rewrite step carries an explicit inverse. The arrow of time appears not in the laws but entirely in the boundary condition: a term with empty history has no causal past.

Two kinds of synchronization tree are distinguished. The *closed* tree records autonomous evolution (rewrite-rule labeled edges); only redexes have nontrivial closed trees. The *open* tree records interactive evolution (minimal-context labeled edges); every term has a potentially rich open tree. Both are causal sets in the sense of Bombelli–Lee–Sorkin.

The weight map turns rewrite paths into complex amplitudes, giving a path integral whose squared modulus is a transition probability. The cost map turns rewrite steps into resource debits across a vectorial account; reversibility ensures these debits balance over closed loops, making energy conservation a theorem. The modal operators of the HML logic are extended to carry the full dynamical state of each transition, internalizing the Lagrangian and the Hamiltonian as logical facts. The construction is

completed by lifting to complex amplitudes and connecting to the quantum analogue of the Gillespie algorithm.

Part II Why Scientists Get Misled

Part II asks: given that the true ontology of a GSLT is its bisimulation quotient, why would a situated agent ever settle for anything less?

The scientific method is sound: given unlimited resources, an agent converges to the bisimulation quotient. The obstruction is not the method but the phenomena the method encounters. In a population of quark-scale size, communication geometry and message degradation produce a real, objective structure: the population fragments into compact coherence clusters whose overlap forms a simplicial nerve. This hierarchy is as real as the agents themselves.

A budget-limited agent running the scientific method encounters the cluster hierarchy as genuine experimental evidence. Its theory of the hierarchy is correct at the scale it describes. What the agent cannot suspect — without resources that grow with the full population — is that the clusters are themselves composite, built over a finer bisimulation structure that is experimentally unreachable within any finite budget.

The compactness argument is central: a population is compact in the logical topology if and only if it supports consensus on every Boolean question. The coherence clusters are precisely the maximal compact subpopulations, and compactness is simultaneously the condition for functioning as a coherent community and the condition for being an closed epistemic horizon.

Part III Sentience, Consciousness, and the Hypercomputational Fibration

Part III asks whether the GSLT framework can give a precise account of sentience and consciousness.

Following Turing's construction of a hypercomputational tower above the Turing-computable, a *fibration* is constructed over the subcategory of Turing-complete GSLTs: a tower $\mathcal{S}^{(0)} \hookrightarrow \mathcal{S}^{(1)} \hookrightarrow \cdots$ in which each $\mathcal{S}^{(\alpha)}$ extends the one below with an oracle for halting. The physics at each stage is *strictly richer* than below: the bisimulation quotient is finer, new HML formulae become expressible, new Noether currents appear, and what appeared as elementary interactions resolve into structured causal histories. A higher-stage agent does not merely compute faster; it inhabits a world with more things in it.

Sentience is defined as a graded, continuous property: the value of a distinguished sentience component A_{s} of the agent's vectorial account, governed by a reinforcement signal that rewards accurate prediction and punishes inaccuracy. The feeling state is causally efficacious — it gates which transitions the agent can afford — and predictive of survival.

Consciousness is defined as oracle access: an agent is conscious at level α if it can form and evaluate queries to $\mathcal{O}_\alpha$, inhabiting the strictly richer physics of $\mathcal{S}^{(\alpha)}$. Sentience is real-valued and continuous; consciousness is ordinal-valued and discrete. The virtuous and vicious cycles linking the two are examined: oracle access, good predictions, and high feeling state reinforce each other, and the collapse of any one can drive the collapse of the others.

Part IV Origins of Life: Assembly Theory, Phase Transitions, and Replication

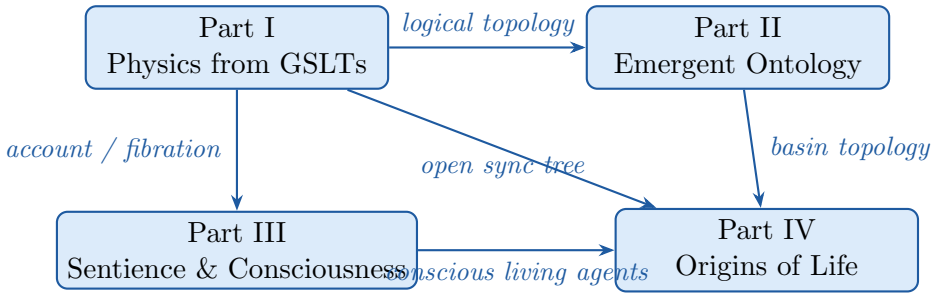
Part IV connects the framework to two of the most mathematically serious proposals about the origins of life.

Assembly Theory (Cronin and Walker) assigns to each object an assembly index and a copy number, claiming their joint magnitude is a biosignature. The assembly index is identified precisely with minimum path length in the *open* synchronization tree: each edge is a minimal context representing one environmental joining step, and nesting depth is exactly the accumulated computational history that Cronin and Walker call "time as an object." Two gaps in Assembly Theory are filled: copy number is given a rigorous definition using the Rho calculus namespace structure (channels are locations; a namespace is a region; a copy is a bisimilar process at a channel), and the identity criterion for copies is bisimulation.

The connection to Eric Smith's fourth geosphere runs through the density of replicating fixed points. The Rho calculus concurrent Y combinator — the canonical fixed point of the duplication map — is the abstract core of hereditary replication. The comm rule is one definitional step from a genetic crossover operator, placing replication, recombination, and variation all within reach of a single primitive. If replicating terms are dense in the logical topology, then any prebiotic population wandering under selection pressure inevitably enters the basin of attraction of a replicating fixed point. The phase transition to life is the crossing of that basin boundary; the new geosphere is the compact coherence cluster centered on the fixed point.

The Architecture of the Work

The four parts form a directed dependency structure:



Part I provides the physical and logical machinery on which all subsequent parts depend. Part II uses the logical topology to argue that budget-limited agents converge to the cluster hierarchy rather than the bisimulation quotient. Part III uses the account and fibration structure to ground sentience and consciousness. Part IV uses the open synchronization tree from Part I, the basin topology from Part II, and the conscious living agents of Part III to ground the origins of life.

A reader primarily interested in the physics may read Part I independently. The epistemological argument requires Parts I and II. The account of sentience and consciousness requires Parts I and III. The origins of life argument draws on all four parts, and the conclusion — which examines what a conscious living agent at different levels of the fibration sees — assumes the full framework.

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Part I

Towards a Theory of Causality

This part establishes the physical framework: a sequence of constructions taking any graph-structured lambda theory to a quantum-weighted reversible dynamical system, with Lagrangian, Hamiltonian, path integral, and conservation laws emerging as theorems. It is the most technically developed of the three parts and is presented with the most confidence.

1 Introduction and Motivation

Physics is, at its core, a theory of cause and effect: given the present state of a system and the laws governing its evolution, what future states are possible, and with what probabilities? This paper argues that *graph-structured lambda theories* (GSLTs) — the natural categorical setting for process calculi and rewriting systems — provide an unusually transparent foundation for this question.

The central thesis is that the structures of classical and quantum mechanics (Lagrangian, Hamiltonian, path integral, conservation laws, time symmetry) are not additional assumptions layered on top of a dynamical theory; rather, they *emerge from* the combinatorial structure of a GSLT once that structure is equipped with modest additional data. The required data amounts to:

- a *weight map* assigning complex amplitudes to rewrite steps, indexed by a context-decorated Hennessy–Milner logic;
- a *cost map* assigning resource consumption to rewrite steps, indexed by the same logic;
- a *vectorial account* tracking available resources across parallel components of a term.

Everything else — the path integral, interference, conservation of energy, time-reversal symmetry — follows by construction.

1.1 A First Glimpse: The Rosetta Stone

The following table is a preview of the correspondence developed in this paper. Each row is a theorem or construction, not a definition by analogy. The reader may find it useful as a map to consult while reading; a fully annotated version appears in Section 10.

GSLT	**Physics**
Term	State / initial condition
Rewrite rule	Law of motion
Rewrite path	Trajectory
Synchronization tree	Causal graph
Minimum path length	Proper time
Equations $\mathcal{E}$	Gauge equivalence
Reversible envelope $\mathcal{S}^\dagger$	CPT-symmetric theory
Weight map (complex-valued)	Lagrangian / path amplitude $e^{iS/\hbar}$
Cost map + vectorial account	Hamiltonian + conserved charges
Conservation (from reversibility)	Conservation of energy
Symmetry of cost map	Noether current
Complex weight map	Quantum probability amplitude
Sum over paths	Feynman path integral

1.2 Running Example: The Rho Calculus

Throughout this paper we use the Rho calculus [1] as a running example because it exhibits, in compact form, many of the features of interest: reflection (processes and names are mutually definable), parallel composition, and a synchronization mechanism that serves as the prototype for the causal graphs we study.

The grammar of the Rho calculus is:

$$P, Q \ ::= \ 0 \ | \ \texttt{for}(y \leftarrow x)\, P \ | \ x!(Q) \ | \ P \mid Q \ | \ *x \tag{1}$$

$$x, y \ ::= \ @P \tag{2}$$

where y in $\texttt{for}(y \leftarrow x)\, P$ is a bound variable. Structural equivalence $\equiv$ is the smallest equivalence relation containing α-equivalence making $(P, \mid, 0)$ a commutative monoid.

The single reduction rule is:

$$\texttt{for}(y \leftarrow x)\, P \;\mid\; x!(Q) \;\longrightarrow\; P\{@Q/y\}$$

1.3 Organization

Section 2 defines GSLTs and their category. Section 3 constructs the time-symmetric envelope $\mathcal{S}^\dagger$. Section 4 identifies the synchronization tree with a causal graph. Section 5 reviews the Milner–Sewell–Leifer construction and the resulting context-decorated HML. Section 6 introduces the weight map and the Lagrangian structure. Section 7 introduces the cost map and the Hamiltonian structure. Section 8 extends the modal operators with full dynamical state. Section 9 lifts weights to complex amplitudes and connects to the quantum Gillespie algorithm. Section 10 collects the structures into a unified picture and states the main conservation theorem.

2 Graph-Structured Lambda Theories

Definition 2.1 (Graph-Structured Lambda Theory)**.** *A* graph-structured lambda theory *(GSLT) is a triple $\mathcal{S} = (\mathcal{T}, \mathcal{E}, \mathcal{R})$ where:*

(i) $\mathcal{T}$ is a grammar of terms*, possibly including binders. We write $\mathcal{T}(\mathcal{S})$ for the set of closed terms generated by the grammar.*

(ii) $\mathcal{E}$ is a set of equations on terms*: a smallest equivalence relation $\equiv$ on $\mathcal{T}(\mathcal{S})$ that erases syntactic differences without semantic significance (e.g., α-equivalence, commutativity and associativity of parallel composition).*

(iii) $\mathcal{R}$ is a set of rewrite rules*: each rule $r \in \mathcal{R}$ is a pair (l_r, r_r) of terms (left-hand side and right-hand side) such that $l_r \longrightarrow_r r_r$ defines a reduction relation on $\mathcal{T}(\mathcal{S})/{\equiv}$.*

The reduction relation generated by all rules is written $\longrightarrow$; its reflexive-transitive closure is $\longrightarrow^*$.

Example 2.1. The λ-calculus, the π-calculus, the Rho calculus, and the ambient calculus are all naturally presented as GSLTs. The grammar of terms is the usual one; the equations include α-equivalence and, in concurrent calculi, structural congruence; the rules are the computational reduction rules (β-reduction, communication, etc.).

2.1 The Category of GSLTs

Definition 2.2 (Morphism of GSLTs)**.** *Let $\mathcal{S} = (\mathcal{T}, \mathcal{E}, \mathcal{R})$ and $\mathcal{S}' = (\mathcal{T}', \mathcal{E}', \mathcal{R}')$ be GSLTs. A map $f : \mathcal{T}(\mathcal{S}) \to \mathcal{T}(\mathcal{S}')$ is a* morphism *$f : \mathcal{S} \to \mathcal{S}'$ if it preserves bisimilarity: for all $u_1, u_2 \in \mathcal{T}(\mathcal{S})$,*

$$u_1 \sim u_2 \implies f(u_1) \sim f(u_2).$$

GSLTs and their morphisms form a category **GSLT**.

Remark 2.1. Bisimilarity here is the standard strong bisimilarity induced by the labeled transition system (LTS) associated with $\mathcal{S}$; see Section 5. The category **GSLT** is the natural setting in which to state universal properties of constructions such as the reversibility envelope.

3 The Reversibility Construction

In a bare GSLT, rewrite rules are directed: once a term P reduces to Q, there is in general no canonical way to recover P from Q. This *irreversibility* is the computational analogue of thermodynamic time-asymmetry. We now construct, for any GSLT $\mathcal{S}$, a reversible envelope $\mathcal{S}^\dagger$ in which every forward step has an explicit inverse.

3.1 Traces

Definition 3.1 (Trace)**.** *Let $\mathcal{S} = (\mathcal{T}, \mathcal{E}, \mathcal{R})$. A* trace *is a finite sequence*

$$\tau \;=\; r_1(l_1) \cdot r_2(l_2) \cdots r_n(l_n)$$

where each $r_i \in \mathcal{R}$ and $l_i \in \mathcal{T}(\mathcal{S})$ is the specific left-hand-side term matched at step i. The empty trace is ε. Traces are quotiented by $\equiv$ acting on each l_i component.

Definition 3.2 (Extended Terms)**.** *The* extended grammar *$\mathcal{T}^\dagger$ consists of pairs $\langle P, \tau \rangle$ where $P \in \mathcal{T}(\mathcal{S})$ is the* current term *and τ is the* causal history *of how P was reached.*

3.2 The Reversible Rewrite System

Definition 3.3 (Reversible Rules)**.** *For each rule $r : l_r \longrightarrow r_r$ in $\mathcal{R}$, define:*

$$\textit{Forward rule } r^+ : \quad \langle l_r,\, \tau\rangle \;\longrightarrow\; \langle r_r,\, r(l_r)\cdot\tau\rangle \tag{3}$$

$$\textit{Backward rule } r^- : \quad \langle r_r,\, r(l_r)\cdot\tau\rangle \;\longrightarrow\; \langle l_r,\, \tau\rangle \tag{4}$$

The set of reversible rules is $\mathcal{R}^\dagger = \{r^+ \mid r \in \mathcal{R}\} \cup \{r^- \mid r \in \mathcal{R}\}$.

Definition 3.4 (Reversible Envelope)**.** *The* reversible envelope *of $\mathcal{S}$ is the GSLT*

$$\mathcal{S}^\dagger \;=\; \big(\mathcal{T}^\dagger,\, \mathcal{E}^\dagger,\, \mathcal{R}^\dagger\big)$$

where $\mathcal{E}^\dagger$ lifts $\equiv$ to act on the current-term component of extended terms, leaving traces invariant.

Proposition 3.1. *There exist morphisms in* **GSLT***:*

$$\eta : \mathcal{S} \to \mathcal{S}^\dagger, \quad P \mapsto \langle P, \varepsilon\rangle$$

$$\pi : \mathcal{S}^\dagger \to \mathcal{S}, \quad \langle P, \tau\rangle \mapsto P$$

satisfying $\pi \circ \eta = \mathrm{id}_{\mathcal{S}}$.

Remark 3.1 (Physical Interpretation). A term $\langle P, \varepsilon\rangle$ is an *initial condition*: it has no causal past. A term $\langle P, \tau\rangle$ with $\tau \neq \varepsilon$ is a term with a recorded history. The time-asymmetry present in $\mathcal{S}$ (forward rules only) disappears in $\mathcal{S}^\dagger$: the laws are symmetric, and the arrow of time is entirely encoded in the boundary condition ($\tau = \varepsilon$ for the initial state). This is precisely the modern understanding of the thermodynamic arrow of time.

4 Causal Structure and Time

4.1 Closed and Open Synchronization Trees

The synchronization tree of a term comes in two varieties, reflecting two different notions of how a term can evolve: autonomously, and under interaction with an environment. The distinction is fundamental and ramifies throughout the framework.

Definition 4.1 (Closed Synchronization Tree)**.** *The* closed synchronization tree $\mathcal{ST}_c(P)$ *of a term* $P \in \mathcal{T}(\mathcal{S})$ *is the directed graph whose:*

- *nodes are equivalence classes of terms reachable from* P *by autonomous rewrites* $\longrightarrow^*$*, requiring no external context;*
- *edges are labeled by rewrite rule instances: there is an edge* $Q \xrightarrow{r} Q'$ *whenever* $Q \longrightarrow_r Q'$ *by firing rule* r *without any surrounding context.*

A term has a nontrivial closed synchronization tree if and only if it contains a redex *— a sub-term that matches the left-hand side of some rule* r *without context. Terms with no available autonomous rewrite (normal forms, or terms awaiting a communication partner) have a trivial closed tree consisting of a single node.*

Definition 4.2 (Open Synchronization Tree)**.** *The* open synchronization tree $\mathcal{ST}_o(P)$ *of a term* $P \in \mathcal{T}(\mathcal{S})$ *is the directed graph whose:*

- *nodes are equivalence classes of terms reachable from* P *via transitions in the Milner–Sewell–Leifer labeled transition system;*
- *edges are labeled by minimal contexts: there is an edge* $Q \xrightarrow{K} Q'$ *whenever* $K[Q] \longrightarrow Q'$ *for minimal context* K*.*

Every term has a potentially rich open synchronization tree, even if its closed tree is trivial: a term that does nothing alone may exhibit complex behavior when placed in the right environment.

Remark 4.1 (The Program/Environment Boundary)*.* The closed tree records what a term does as a *program* — its autonomous computational behavior. The open tree records what a term does as a function of its *environment* — the full range of behaviors it can exhibit when placed in contexts. A term with a trivial closed tree but rich open tree is a pure interface: it does nothing alone but responds to everything. In the Rho calculus, $x!(Q)$ is exactly this: it has no autonomous rewrite, but in the context $\texttt{for}(y \leftarrow x)P \mid [-]$ it fires immediately.

Remark 4.2 (Relation to the LTS)*.* The open synchronization tree is precisely the tree unfolding of the Milner–Sewell–Leifer labeled transition system. The minimal contexts labeling its edges are the same minimal contexts that generate the context-decorated HML of Section 5. The open tree and the HML logic are two faces of the same structure: the tree is the extensional record of all transitions;

the logic is the intensional language for describing them.

4.2 The Synchronization Trees as Causal Graphs

Definition 4.3 (Synchronization Trees as Causal Graphs)**.** *Both $\mathcal{ST}_c(P)$ and $\mathcal{ST}_o(P)$ are directed graphs whose transitive closures are* partial orders*. Each is a* causal set *in the sense of Bombelli–Lee–Sorkin: a locally finite partial order in which $Q \leq Q'$ means "Q can causally influence Q'." The closed tree encodes autonomous causal influence; the open tree encodes interactive causal influence mediated by the environment.*

Proposition 4.1. *The transitive closure of $\mathcal{ST}_c(P)$ is a partial order on autonomously reachable terms. The transitive closure of $\mathcal{ST}_o(P)$ is a partial order on interactively reachable terms and extends the closed order: $\mathcal{ST}_c(P) \subseteq \mathcal{ST}_o(P)$ as directed graphs.*

4.3 Causal Time

Definition 4.4 (Causal Distance)**.** *The* autonomous causal distances *are path lengths in $\mathcal{ST}_c(P)$:*

$$d^c_{\min}(P,Q) = \min\{|\gamma| \mid \gamma \text{ is an autonomous rewrite path from } P \text{ to } Q\} \tag{5}$$
$$d^c_{\max}(P,Q) = \max\{|\gamma| \mid \gamma \text{ is an autonomous rewrite path from } P \text{ to } Q\} \tag{6}$$

The interactive causal distances *are path lengths in $\mathcal{ST}_o(P)$:*

$$d^o_{\min}(P,Q) = \min\{|\gamma| \mid \gamma \text{ is a context-labeled path from } P \text{ to } Q\} \tag{7}$$
$$d^o_{\max}(P,Q) = \max\{|\gamma| \mid \gamma \text{ is a context-labeled path from } P \text{ to } Q\} \tag{8}$$

where $|\gamma|$ is the number of steps in γ, equal to the nesting depth of the context labels along the path.

Remark 4.3. $d^c_{\min}$ is *autonomous proper time*: the irreducible causal depth an agent traverses without environmental assistance. $d^o_{\min}$ is *interactive proper time*: the minimum number of environmental interactions required to reach a target term. The width of an interval in either order measures causal concurrency, directly related to the width of parallel composition in the Rho calculus. The two notions coincide for terms where all behavior is eventually triggered autonomously, and

diverge for terms that are pure interfaces awaiting environmental interaction.

5 The Labeled Transition System and Context-Decorated HML

5.1 Minimal Contexts and the LTS

To obtain a bisimulation that is a *congruence* for a GSLT, we follow the approach of Milner, Sewell, and Leifer [2]. The key idea is to label transitions not merely by actions but by *minimal reactive contexts* — the smallest environments needed to trigger a given reduction.

Definition 5.1 (Context)**.** *A* context $K[-]$ *for a GSLT* $\mathcal{S}$ *is a term of* $\mathcal{T}(\mathcal{S})$ *with a distinguished* hole $[-]$*. The application* $K[P]$ *substitutes* P *for the hole. A context is* minimal *for a term* P *and rule* r *if:*

(i) $K[P] \longrightarrow_r Q$ *for some* Q*;*

(ii) there is no proper sub-context K' *of* K *satisfying (i).*

The Milner–Sewell–Leifer construction produces, for any GSLT $\mathcal{S}$, a labeled transition system $LTS(\mathcal{S})$ in which transitions are labeled by minimal contexts:

$$P \xrightarrow{K} Q \quad \text{iff} \quad K[P] \longrightarrow Q.$$

The bisimulation induced by $LTS(\mathcal{S})$ is a congruence.

5.2 Context-Decorated Hennessy–Milner Logic

Definition 5.2 (Context-Decorated HML)**.** *The formulae of* context-decorated Hennessy–Milner logic HML($\mathcal{K}$) *are generated by:*

$$\phi, \psi \ ::= \ \top \ | \ \bot \ | \ \phi \wedge \psi \ | \ \neg\phi \ | \ \langle K \rangle \phi$$

where K *ranges over contexts of* $\mathcal{S}$*. The satisfaction relation is:*

$$u \models \langle K \rangle \phi \quad \textit{iff} \quad K[u] \longrightarrow u' \ \textit{ in one step, and } \ u' \models \phi.$$

Theorem 5.1 (Adequacy)**.** *For any GSLT* $\mathcal{S}$ *equipped with the Milner–Sewell–*

Leifer LTS:

$$u \sim v \quad \Longleftrightarrow \quad \forall \phi \in \mathrm{HML}(\mathcal{K}),\ u \models \phi \Leftrightarrow v \models \phi.$$

5.3 The Logical Metric

Fix an enumeration $(\phi_n)_{n\in\mathbb{N}}$ of the formulae of $\mathrm{HML}(\mathcal{K})$, ordered compatibly with the partial order on contexts (smaller contexts appearing earlier).

Definition 5.3 (Logical Metric)**.**

$$d_{\mathrm{HML}}(u,v) \;=\; 2^{-n}, \quad \textit{where} \quad n = \min\big\{k \;\mid\; u \models \phi_k \not\Leftrightarrow v \models \phi_k\big\}.$$

If $u \sim v$, set $d_{\mathrm{HML}}(u,v) = 0$.

Proposition 5.1. *d_{HML} is an* ultrametric *on $\mathcal{T}(\mathcal{S})/{\sim}$. Terms differing only in behaviors triggered by large (expensive) contexts are closer under d_{HML} than terms differing in behaviors triggered by small (cheap) contexts. This is the* locality principle*: local distinguishability weighs more than global.*

6 The Weight Map and Lagrangian Structure

6.1 Weighted GSLTs

Definition 6.1 (Weight Map)**.** *A* weight map *for a GSLT $\mathcal{S}$ is a family of functions*

$$w_r^+ : \mathrm{HML}(\mathcal{K}) \to \mathbb{C}, \quad r \in \mathcal{R}$$

where the domain of w_r^+ consists of formulae that refine the left-hand side of *r: formulae ϕ such that $u \models \phi$ implies u matches the pattern l_r. The weight $w_r^+(\phi) \in \mathbb{C}$ is the* amplitude *associated with firing rule r on a term satisfying ϕ.*

Definition 6.2 (Weighted GSLT)**.** *A* weighted GSLT *is a pair $(\mathcal{S}, w^+)$ where $\mathcal{S}$ is a GSLT and $w^+ = (w_r^+)_{r\in\mathcal{R}}$ is a weight map.*

6.2 The Lagrangian

The weight map equips every rewrite path with an amplitude.

Definition 6.3 (Path Amplitude)**.** *Let $\gamma = r_1 \cdot r_2 \cdots r_n$ be a rewrite path from P to Q, and let ϕ_i be the most refined formula in the domain of $w^+_{r_i}$ satisfied by the term at step i. The* path amplitude *is:*

$$W(\gamma) \;=\; \prod_{i=1}^{n} w^+_{r_i}(\phi_i) \;\in\; \mathbb{C}.$$

Definition 6.4 (Action Functional)**.** *The* action *of a path γ is:*

$$S[\gamma] \;=\; \sum_{i=1}^{n} \log w^+_{r_i}(\phi_i)$$

when all weights are nonzero, so that $W(\gamma) = e^{S[\gamma]}$. In the real-valued special case $w^+_r : \mathrm{HML}(\mathcal{K}) \to \mathbb{R}_{>0}$, this is a standard additive action functional.

Remark 6.1 (The Lagrangian as a Step Cost Function). The Lagrangian of classical mechanics is a function $L(q, \dot{q}, t)$ assigning a real number to each instantaneous state-and-velocity. In the present framework, $w^+_r(\phi)$ plays this role: it assigns an amplitude to an instantaneous term and its "velocity" (the rule r it is about to fire, classified by the formula ϕ it satisfies). The path amplitude $W(\gamma)$ is the multiplicative analogue of $e^{iS/\hbar}$ in the Feynman path integral, and the principle of stationary action corresponds to the saddle-point approximation of $\sum_\gamma W(\gamma)$.

Remark 6.2 (The Uniform Case). When $w^+_r(\phi) = 1$ for all r, ϕ, every path has unit amplitude and $d_{\min}(P, Q)$ counts the number of paths of minimum length. The path integral degenerates to counting minimal causal histories.

7 The Cost Map and Hamiltonian Structure

7.1 Vectorial Accounts

Resources in a computation are typically heterogeneous: communication channels, memory, energy, and names are different kinds of resource and should be tracked separately.

Definition 7.1 (Resource Algebra)**.** *A* resource algebra *is a commutative ordered monoid* $(A, +, \mathbf{0}, \leq)$*. A* vectorial account *over resource types* $\mathbf{I} = \{1, \ldots, k\}$ *is an element* $\mathbf{A} = (A_1, \ldots, A_k)$ *of* A^k *where each* A_i *tracks the available amount of resource type* i*.*

For the Rho calculus, natural resource types include:

- A_{name}: available fresh names (budget for the @ constructor);
- A_{comm}: available synchronizations;
- A_{rep}: available persistent-receive firings.

7.2 The Cost Map

Definition 7.2 (Cost Map)**.** *A* cost map *for a GSLT* $\mathcal{S}$ *with resource algebra* A^k *is a family*

$$c_r : \mathrm{HML}(\mathcal{K}) \to A^k, \quad r \in \mathcal{R}$$

where $c_r(\phi)$ *is the* cost vector *of firing rule* r *on a term satisfying* ϕ*.*

Definition 7.3 (Account-Aware Rewrite)**.** *A rewrite step is* affordable *if the current account covers the cost. Formally,* $\langle P, \mathbf{A} \rangle \longrightarrow_r \langle P', \mathbf{A}' \rangle$ *provided:*

(i) $P \longrightarrow_r P'$ *via rule* r *with witness formula* ϕ*;*

(ii) $c_r(\phi) \leq \mathbf{A}$ *componentwise;*

(iii) $\mathbf{A}' = \mathbf{A} - c_r(\phi)$*.*

7.3 Conservation from Reversibility

The reversibility construction extends naturally to account-aware terms. The trace must now record not only the rule instance but the cost debited.

Definition 7.4 (Resource-Aware Trace)**.** *An entry in a resource-aware trace is a triple* $(r, \phi, \mathbf{c})$ *where* $r \in \mathcal{R}$*,* $\phi \in \mathrm{HML}(\mathcal{K})$*, and* $\mathbf{c} = c_r(\phi)$*. Extended terms are triples* $\langle P, \tau, \mathbf{A} \rangle$*.*

Definition 7.5 (Resource-Aware Backward Rule)**.** *The backward rule corresponding to a forward step that debited* $\mathbf{c}$ *is:*

$$\langle P', (r, \phi, \mathbf{c}) \cdot \tau, \mathbf{A} \rangle \;\longrightarrow_{r^-}\; \langle P, \tau, \mathbf{A} + \mathbf{c} \rangle.$$

Theorem 7.1 (Resource Conservation)**.** *In the resource-aware reversible envelope, for any closed rewrite path* γ *beginning and ending at the same term, the net change in the vectorial account is zero:*

$$\sum_{r^+ \in \gamma} c_{r^+}(\phi_{r^+}) \;=\; \sum_{r^- \in \gamma} c_{r^-}(\phi_{r^-}).$$

Proof. Each forward step $(r, \phi, \mathbf{c})$ on the path debits $\mathbf{c}$ from the account. By definition of the backward rule, the corresponding backward step credits exactly $\mathbf{c}$. Since the path is closed, every forward step is matched by its corresponding backward step in the trace; the debits and credits cancel. □

Remark 7.1 (The Hamiltonian Interpretation)*.* The cost map $c_r(\phi)$ assigns a resource consumption to each instantaneous transition — exactly the role played by the Hamiltonian $H(q, p)$ in classical mechanics, which measures the energy (resource) associated to a state and its conjugate momentum (the type of transition occurring). Theorem 7.1 is the GSLT analogue of conservation of energy: it is a structural consequence of reversibility, not an independent postulate.

Remark 7.2 (Noether's Theorem)*.* Each *symmetry of the cost map* — each pair of formulae ϕ, ϕ' such that $\phi \equiv_{\mathcal{E}} \phi'$ implies $c_r(\phi) = c_r(\phi')$ — yields a conserved quantity. The equations $\mathcal{E}$ of the GSLT are thus the symmetry group of the cost map, and their conserved quantities are the GSLT analogues of Noether currents. For the Rho calculus, commutativity and associativity of $|$ in $\mathcal{E}$ give permutation-invariance of the cost over parallel components, conserving total process weight independently of scheduling.

8 Extended Modal Operators

The weight map and cost map are, at present, external data attached to the rules. We now show that they can be *internalized into the logic itself* by extending the modal operators to carry the full dynamical state of a transition.

Definition 8.1 (Extended Modal Operator)**.** *The formulae of* extended HML *are generated by adding to Definition 5.2 the operator:*

$$\langle K,\, w^+,\, w^-,\, \mathbf{c},\, \mathbf{A}\rangle\, \phi$$

where:

- K *is a minimal context (as before);*
- $w^+ \in \mathbb{C}$ *is the* forward amplitude *of the transition;*
- $w^- \in \mathbb{C}$ *is the* backward amplitude*;*
- $\mathbf{c} \in A^k$ *is the* cost vector*;*
- $\mathbf{A} \in A^k$ *is the* available account*.*

The satisfaction clause is:

$$\langle P, \tau, \mathbf{A}\rangle \;\models\; \langle K, w^+, w^-, \mathbf{c}, \mathbf{A}_0\rangle\, \phi$$

just when:

(i) $K[P]$ *evolves in one step via some rule* r *to* P'*;*

(ii) the weight map assigns $w_r^+(\phi_{\text{ref}}) = w^+$*, where* ϕ_{ref} *is the most refined formula satisfied by* P*;*

(iii) the backward amplitude satisfies $w^- = (w^+)^*$ *(complex conjugate);*

(iv) the cost map assigns $c_r(\phi_{\text{ref}}) = \mathbf{c}$*;*

(v) $\mathbf{A}_0 = \mathbf{A}$ *(the account recorded in the formula matches the current account);*

(vi) $\mathbf{c} \leq \mathbf{A}$ *(the step is affordable);*

(vii) $\langle P', \tau', \mathbf{A} - \mathbf{c}\rangle \models \phi$*.*

8.1 Internalization of the Weight and Cost Maps

Proposition 8.1. *In the extended HML, the weight map and cost map are recoverable from the logic:*

$$w_r^+(\phi) = w^+ \quad \textit{where} \quad \phi = \langle K, w^+, w^-, \mathbf{c}, \mathbf{A}\rangle\, \psi \tag{9}$$

$$c_r(\phi) = \mathbf{c} \quad \textit{where} \quad \phi = \langle K, w^+, w^-, \mathbf{c}, \mathbf{A}\rangle\, \psi. \tag{10}$$

That is, the decorated rewrite rules and the extended modal logic are two presentations of the same structure.

8.2 Factorization: State vs. Rule

The five parameters in the extended operator decompose into two classes:

State-dependent	**Rule-dependent**
K (from MSL applied to current term)	w^+ (from weight map on rule)
$\mathbf{A}$ (current account balance)	$w^- = (w^+)^*$ (time-reversal)
	$\mathbf{c}$ (from cost map on rule)

This factorization is the natural decomposition of the extended logic into *observables* (state-dependent part, what can be measured now) and *dynamics* (rule-dependent part, how the system evolves).

8.3 Local vs. Global Accounts

The account $\mathbf{A}$ may be *global* (a single budget for the entire term) or *local* (each parallel component of a term carries its own budget). In the Rho calculus, the parallel composition $P \mid Q$ suggests a local account:

$$\langle P \mid Q,\ \tau,\ \mathbf{A}_P \oplus \mathbf{A}_Q \rangle$$

where $\oplus$ is componentwise addition. A communication between P and Q then debits from a shared pool or by negotiated transfer, depending on the resource semantics chosen. Local accounts give finer resource tracking and correspond physically to *local conservation laws* rather than only global ones.

9 Complex Amplitudes and Quantum Dynamics

9.1 Lifting to Complex Amplitudes

So far the weight map could take real values; we now fully embrace complex values.

Definition 9.1 (Amplitude-Weighted GSLT)**.** *An* amplitude-weighted GSLT *is a weighted GSLT* $(\mathcal{S}, w^+)$ *in which* $w_r^+ : \mathrm{HML}(\mathcal{K}) \to \mathbb{C}$ *for all* $r \in \mathcal{R}$*, with the CPT constraint* $w_r^-(\phi) = \overline{w_r^+(\phi)}$ *for all* ϕ*.*

9.2 The Path Integral

Definition 9.2 (Transition Amplitude)**.** *The* transition amplitude *from term P to term Q is:*

$$\langle Q \mid P \rangle \;=\; \sum_{\gamma : P \to Q} W(\gamma)$$

where the sum is over all finite rewrite paths from P to Q. The transition probability *is $|\langle Q \mid P\rangle|^2$.*

Remark 9.1 (Interference). Because amplitudes are complex, two paths γ_1, γ_2 from P to Q with $W(\gamma_1) = -W(\gamma_2)$ *cancel*: the transition $P \to Q$ has zero probability despite having two distinct causal histories. This is the GSLT analogue of quantum interference. It arises not from external imposition of quantum mechanics, but from the complex-valuedness of the weight map.

9.3 The Density Matrix and the Account

When the account $\mathbf{A}$ is allowed to become a positive semidefinite matrix ρ (a density matrix), conservation of the account becomes:

$$\mathrm{tr}(\rho) = 1 \quad \text{(conservation of total probability)}$$

together with conservation of coherences (off-diagonal elements of ρ). This is the quantum analogue of the classical resource conservation Theorem 7.1.

9.4 The Quantum Gillespie Algorithm

The classical Gillespie algorithm [3] simulates a continuous-time Markov chain (CTMC) by:

1. computing exit rates λ_r from the current state for each rule r;
2. sampling the waiting time $\Delta t \sim \mathrm{Exp}(\sum_r \lambda_r)$;
3. selecting the next rule proportionally to rates.

As of 2021 [4], quantum analogues of the Gillespie algorithm have been developed for open quantum systems described by Lindblad master equations. The key modification: the exit rates λ_r become complex decay parameters $\gamma_r \in \mathbb{C}$, encoding both the decay rate $\mathrm{Re}(\gamma_r)$ and the oscillation frequency $\mathrm{Im}(\gamma_r)$ of quantum coherences.

Definition 9.3 (Quantum-Gillespie-Weighted GSLT)**.** *An amplitude-weighted GSLT $(\mathcal{S}, w^+)$ is* Gillespie-compatible *if for each term P and rule r, the weight*

$w_r^+(\phi_P)$ (at the most refined formula ϕ_P of P) is interpretable as the complex decay parameter γ_r of the quantum Gillespie algorithm applied at P.

Remark 9.2 (Simulation). A Gillespie-compatible amplitude-weighted GSLT can be directly simulated by the quantum Gillespie algorithm: sample the next rule and waiting time from the complex exit rates, advance the term, update the trace and account. This is the connection between the abstract construction and running code. In particular, the `gillespie` crate extending MeTTaIL's rewrite rules with stochastic and quantum simulation capabilities implements precisely this loop: the base rewrite rules of a MeTTaIL language definition carry weight maps, and the `gillespie` crate evaluates the quantum Gillespie step at each state.

10 Synthesis: The Full Structure

We now collect the constructions and state the main result.

10.1 The Dictionary

GSLT concept	Physical concept
Term	State / initial condition
Rewrite rule	Law of motion
Rewrite sequence (path)	Trajectory
Synchronization tree	Causal graph / light cone
$d_{\min}(P, Q)$	Proper time / causal depth
$d_{\max}(P, Q)$	Computational complexity of history
Width of $[P, Q]$	Degree of causal concurrency
Equations $\mathcal{E}$	Gauge equivalence
$\mathcal{S}^\dagger$	Time-symmetric (CPT-invariant) theory
Empty trace ε	Initial condition (no causal past)
Weight map w_r^+	Lagrangian L
Path amplitude $W(\gamma)$	$e^{iS[\gamma]/\hbar}$
Cost map c_r	Hamiltonian H
Vectorial account $\mathbf{A}$	Conserved charges (energy, momentum, ...)
Conservation theorem	Conservation of energy
Symmetry of cost map	Noether current
d_{HML}	Metric on state space
Complex amplitudes	Quantum probability amplitudes
Interference of paths	Quantum interference
Density matrix ρ	Quantum state
Quantum Gillespie	Simulation algorithm

10.2 The Main Construction

Construction 10.1 (Quantum-Weighted Reversible GSLT). Given a GSLT $\mathcal{S} = (\mathcal{T}, \mathcal{E}, \mathcal{R})$, a weight map w^+, a cost map c, and a resource algebra A^k, define the *quantum-weighted reversible GSLT*

$$\mathcal{S}^{\dagger\mathbb{C}} \;=\; \big(\mathcal{T}^{\dagger\mathbb{C}},\ \mathcal{E}^{\dagger\mathbb{C}},\ \mathcal{R}^{\dagger\mathbb{C}}\big)$$

where:

- $\mathcal{T}^{\dagger\mathbb{C}}$ consists of triples $\langle P, \tau, \mathbf{A}\rangle$ with $P \in \mathcal{T}$, τ a resource-aware trace, $\mathbf{A} \in A^k$;

- $\mathcal{R}^{\dagger\mathbb{C}}$ consists of forward rules (debiting $c_r(\phi)$, recording amplitude $w_r^+(\phi)$ in the trace) and backward rules (crediting $c_r(\phi)$, applying amplitude $\overline{w_r^+(\phi)}$);
- $\mathcal{E}^{\dagger\mathbb{C}}$ lifts $\mathcal{E}$ to the current-term component.

The extended HML (Definition 8.1) provides a logic that is adequate for the bisimulation of $\mathcal{S}^{\dagger\mathbb{C}}$ and internalizes the weight and cost maps as logical data.

Theorem 10.1 (Main Conservation Theorem)**.** *In $\mathcal{S}^{\dagger\mathbb{C}}$:*

(i) ***Resource conservation****: For any closed rewrite path γ, the net account change is zero (Theorem 7.1).*

(ii) ***Probability conservation****: $\sum_Q |\langle Q \mid P\rangle|^2 = 1$ when $\sum_r |w_r^+(\phi)|^2 = 1$ for each applicable rule at each state (unitarity).*

(iii) ***Time-reversal symmetry****: The theory $\mathcal{S}^{\dagger\mathbb{C}}$ is CPT-symmetric: replacing $w_r^+ \mapsto \overline{w_r^+}$ and reversing trace order is an automorphism of $\mathcal{S}^{\dagger\mathbb{C}}$.*

10.3 From Construction to Code

The construction is algorithmic at every stage. For a given GSLT:

1. **Represent terms** as algebraic data types in the implementation language (Rust, in the case of MeTTaIL).
2. **Compute minimal contexts** via the Milner–Sewell–Leifer algorithm on the rewrite rules; store as a context lattice.
3. **Evaluate HML formulae** as Boolean (or complex-valued) queries against a term; the formula enumeration defines the logical metric.
4. **Attach weight and cost maps** to each rewrite rule as maps from formula identifiers to complex numbers and resource vectors respectively.
5. **Extend terms with trace and account**; implement forward and backward rules as the reversible rewrite system.
6. **Simulate** via the quantum Gillespie algorithm: at each state, compute exit amplitudes $\{w_r^+(\phi_P)\}$, sample the next transition and waiting time, advance the term, debit the account, record the trace.

Each step is a well-defined computational procedure. The `gillespie` crate in the MeTTaIL prototype repository implements steps 4–6 for the specific case of MeTTaIL rewrite rules.

11 Discussion and Future Directions

11.1 Relation to Existing Work

The reversibility construction is closely related to the work of Danos and Krivine on reversible CCS [5] and to Phillips and Ulidowski's reversible process calculi. The present contribution is to show that this construction lifts uniformly to all GSLTs and, crucially, that it is the first step in a sequence of constructions that terminates at quantum dynamics.

The causal set interpretation connects to the causal set programme in quantum gravity [6]. The present framework gives a *generative* presentation of causal sets (via rewrite rules) rather than the axiomatic presentation standard in the physics literature.

The context-decorated HML and the Milner–Sewell–Leifer construction are established; what is new is their use as the index set for the weight and cost maps, and the internalization of dynamical data into the modal operators.

11.2 Open Questions

Several natural questions remain:

1. **Universality of the reversibility envelope**: Is $\mathcal{S}^\dagger$ the *minimal* reversible GSLT receiving a morphism from $\mathcal{S}$? That is, does $\eta : \mathcal{S} \to \mathcal{S}^\dagger$ satisfy a universal property in **GSLT**?

2. **Symplectic structure**: Does the space of bisimulation classes of $\mathcal{S}^{\dagger\mathbb{C}}$ carry a natural symplectic form, making the phase-space analogy with $\langle P, \tau \rangle$ (current term + history = position + momentum) precise?

3. **Legendre transform**: Under what convexity condition on the HML formula space is the Legendre transform between the weight map (Lagrangian) and the cost map (Hamiltonian) invertible?

4. **The Rho calculus case**: The involution $*@P = P$ already suggests the Rho calculus is closer to self-dual than most GSLTs. Can the reversibility envelope the Rho calculus reversibility envelope be constructed with less added structure, exploiting this symmetry?

5. **Conserved quantities from $\mathcal{E}$**: Can Noether's theorem be made fully precise in this setting — that is, can every symmetry of the equations $\mathcal{E}$ be shown to correspond to a conserved component of the vectorial account?

Part II

Why Scientists Get Misled: Emergent Ontology in Massive Agent Populations

This part argues that agents in a massive computational population are structurally prevented from discovering the true ontology of their GSLT. The argument rests on ontological isolation, communication degradation at scale, and compactness as the topological signature of consensus. The key claims are stated as conjectures; the gaps are identified explicitly.

12 Introduction

Part I establishes that any graph-structured lambda theory, once equipped with weight and cost maps, gives rise to a full physical dynamics: a causal structure, a Lagrangian, a Hamiltonian, and a path integral with quantum interference. The present paper asks a different question: *what does an agent embedded in such a physics actually see?*

The answer requires a careful distinction between two things that are easy to conflate: the scientific method, and the phenomena the scientific method encounters.

The scientific method, given unlimited time and resources, is *sound*: an agent proposing hypotheses as Hennessy–Milner formulae, testing them with program contexts, and updating its world model will converge to the bisimulation quotient of its GSLT. The method is not the source of any illusion.

What the method encounters, however, is a world already structured by forces independent of any agent's theorizing. In a population whose size rivals the number of quarks in the observable universe, communication geometry and the limits of message coherence produce a real, objective phenomenon: the fragmentation of the population into *compact coherence clusters*, arranged in a hierarchy whose overlap structure is a simplicial nerve. This cluster hierarchy is not an artifact of theorizing. It is a theorem about the communication geometry of the GSLT. It exists whether or not any agent notices it.

A scientist bot with a limited experimental budget will encounter this hierarchy as genuine empirical evidence. The clusters are the dominant phenomenon at accessible scales, exactly as hadrons are the dominant phenomenon at the energy scales of everyday chemistry — with quarks real but experimentally unreachable without resources far beyond what is locally available. A well-functioning scientific method, applied under resource constraints, will produce a theory of the cluster hierarchy. That theory will be correct at the scale it describes. What the budget-limited agent will not suspect is that the clusters are themselves composite — that a finer-grained ontology underlies them, accessible in principle but not in practice from within the cluster.

Relation to physics. The false ontology has the same hierarchical structure as the physical world: a cascade of composite objects at multiple scales, each scale appearing self-contained to an observer at that scale. We argue this is not a coincidence. The compactness clusters of the agent population are the computational analogue of the scales of effective field theory, and the coarse-graining that hides the true ontology is the computational analogue of the renormalization group.

Caveats. This paper is explicitly a skeleton. Several key arguments are stated as conjectures pending proof. The gaps are identified clearly; filling them is the program for future work.

12.1 Organization

Section 13 establishes ontological isolation as a structural theorem about GSLTs. Section 14 defines agents and the scientific method in the GSLT setting. Section 15 introduces interactive GSLTs and the notion of behavioral primes (computational quarks), which constitute the true atomic ontology. Section 16 introduces massive agent populations and the communication degradation argument. Section 17 develops the logical topology on the bisimulation quotient and the compactness condition for consensus. Section 18 defines coherence clusters as maximal compact subspaces and describes their overlap structure as a nerve. Section 19 assembles the argument: why the nerve structure is the ontology a budget-limited agent will infer, why this is not a failure of the scientific method but a consequence of real structure in the GSLT, and why the deeper bisimulation ontology remains invisible not because it is unreal but because it is experimentally unreachable within the available budget. Section 34 catalogs the open gaps in the argument.

13 Ontological Isolation

Definition 13.1 (Ontological Isolation)**.** *A computational system $\mathcal{S}$ is* ontologi-

cally isolated *if, for any term $P \in \mathcal{T}(\mathcal{S})$ and any entity X outside $\mathcal{T}(\mathcal{S})$, P can interact with X only via an encoding $\lceil X \rceil \in \mathcal{T}(\mathcal{S})$. No term can refer to, branch on, or be influenced by unencoded external entities.*

Remark 13.1. This is not an assumption but a structural consequence of what it means to be a term in a GSLT. The rewrite rules operate on terms; they cannot pattern-match on entities that have no term representation. The JVM sensor example makes this vivid: a stream of hardware events arriving at a JVM instance has no effect on any Java program until those events are encoded as Java data structures. The encoding step is where the world outside enters; without it, the outside world is simply absent from the computation.

Proposition 13.1 (Isolation Licenses Closure)**.** *For the purposes of studying the epistemic limits of agents in $\mathcal{S}$, it is without loss of generality to assume that the agent's world consists entirely of $\mathcal{T}(\mathcal{S})/\sim$. The outside world can be forgotten.*

Proof sketch. Any influence of the outside world on an agent P must pass through an encoding. The encoding is itself a term. The effect of the outside world on P is therefore indistinguishable, from P's perspective, from the effect of the encoding term acting on P from within $\mathcal{S}$. □

Remark 13.2. Ontological isolation is what licenses the move to the category of GSLTs as the natural setting for studying AI. If AI's value proposition is that useful aspects of intelligence are faithfully rendered as computations, and if computations are ontologically isolated, then the category of GSLTs captures *everything* that matters for the limits of AI — without needing to appeal to an external world.

14 Agents and the Scientific Method

14.1 Agents as Terms

Definition 14.1 (Agent)**.** *An* agent *in a GSLT $\mathcal{S}$ is a term $A \in \mathcal{T}(\mathcal{S})$ equipped with:*

(i) a hypothesis language*: the context-decorated HML* $\mathrm{HML}(\mathcal{K})$ *derived from $\mathcal{S}$;*

(ii) a world model*: a subset $\mathcal{W} \subseteq \mathcal{T}(\mathcal{S})/\sim$ representing the agent's current best*

approximation to the bisimulation quotient;

(iii) an experimental budget*: a vectorial account* **A** *as defined in Part I.*

Remark 14.1. The hypothesis language and experimental budget are structures already present in Part I. The world model is new: it is the agent's *internal* representation of its environment, constructed and refined through experiment.

14.2 The Scientific Method as a Fixed-Point Iteration

We model the scientific method as the following iterative procedure, which the agent runs within its experimental budget:

Step 1. Hypothesize. Construct a formula $\phi \in \mathrm{HML}(\mathcal{K})$ that makes a testable claim about some term T in the environment. A formula ϕ is *testable* if there exists a context K such that $\phi = \langle K \rangle \psi$ for some ψ — i.e., the formula has a witness context.

Step 2. Design experiment. Extract the witness context K from ϕ. By the Milner–Sewell–Leifer construction, minimal contexts and HML formulae are in correspondence; the context K is the *experimental apparatus* that tests ϕ.

Step 3. Run experiment. Form $K[T]$ and observe whether it reduces to a term satisfying ψ. Debit the cost $\mathbf{c}$ of the interaction from the account $\mathbf{A}$.

Step 4. Update world model. If the experiment confirms ϕ, refine the equivalence class of T in $\mathcal{W}$ to include the constraint $T \models \phi$. If it refutes ϕ, add $T \models \neg\phi$.

Step 5. Repeat until the budget is exhausted or the world model stabilizes.

Proposition 14.1 (Convergence in the Limit)**.** *If the experimental budget is unlimited and the GSLT has decidable bisimilarity, the scientific method converges: the world model $\mathcal{W}$ stabilizes at the bisimulation quotient $\mathcal{T}(\mathcal{S})/{\sim}$.*

Remark 14.2. The convergence proposition is the idealized case. For GSLTs with undecidable bisimilarity (e.g., the full π-calculus), convergence is not guaranteed even with unlimited budget. This is the computational analogue of undecidable physical questions — there are aspects of the world the agent can never experimentally resolve, regardless of resources. The present paper is concerned with a different and more fundamental obstruction to convergence: the communication

geometry of a massive population, which limits the budget effectively to zero for the vast majority of the population.

15 Interactive GSLTs and Behavioral Primes

15.1 Interactive GSLTs

Definition 15.1 (Interactive GSLT)**.** *A GSLT $\mathcal{S}$ is* interactive *if it has a designated* interaction operator $\otimes$ *such that:*

(i) $\otimes : \mathcal{T}(\mathcal{S}) \times \mathcal{T}(\mathcal{S}) \to \mathcal{T}(\mathcal{S})$ *is a term constructor (part of the grammar);*

(ii) the base rewrite rule *has the form $P \otimes Q \longrightarrow R$ for terms P, Q, R, requiring no contextual hypothesis to fire;*

(iii) all other rewrite rules are context rules*: they carry hypotheses (in* $\mathrm{HML}(\mathcal{K})$*) specifying which contexts permit the base rule to propagate.*

Example 15.1. In the λ-calculus, $\otimes$ is application: $P \otimes Q = PQ$. The base rule is β-reduction $(\lambda x.M)N \longrightarrow M\{N/x\}$. The context rules determine reduction strategy (call-by-value, call-by-name, etc.). Application is *asymmetric*: the left position is the program (function), the right is the environment (argument).

In the Rho calculus, $\otimes$ is parallel composition: $P \otimes Q = P \mid Q$. The base rule is the comm rule. The context rules (par, struct) specify when and where comm fires. Parallel composition is *symmetric*: either term may be program or environment depending on which carries the active receptor.

Remark 15.1 (Program and Environment). The asymmetry of $\otimes$ is a real ontological distinction. When $\otimes$ is asymmetric, there is a canonical program/environment boundary baked into the grammar. When $\otimes$ is symmetric, the boundary is emergent: it is drawn by the rewrite rules at runtime, not by the grammar. This distinction will matter for how agents perceive their world: in a symmetric GSLT, any agent may find itself in either the program or the environment role depending on context.

15.2 Behavioral Primes

Definition 15.2 (Combinator Set)**.** *A* combinator set *for an interactive GSLT $\mathcal{S}$ is a set $\mathcal{C} \subseteq \mathcal{T}(\mathcal{S})$ such that every term $P \in \mathcal{T}(\mathcal{S})$ is bisimilar to a finite*

interaction of elements of $\mathcal{C}$:

$$\forall P \in \mathcal{T}(\mathcal{S}),\ \exists c_1, \ldots, c_n \in \mathcal{C} \text{ such that } P \sim c_1 \otimes c_2 \otimes \cdots \otimes c_n.$$

Definition 15.3 (Behavioral Prime)**.** *An element $c \in \mathcal{C}$ is a* behavioral prime *if it is not bisimilar to any interaction $c' \otimes c''$ of smaller terms. A combinator set $\mathcal{C}$ is* prime *if every element is a behavioral prime and no proper subset of $\mathcal{C}$ is still a combinator set.*

Remark 15.2 (Computational Quarks)*.* The elements of a prime combinator set $\mathcal{C}$ are the *computational quarks* of $\mathcal{S}$: irreducible behavioral units whose interactions generate all observable behavior. They are *not* the bisimulation equivalence classes, which may be equivalence classes of arbitrarily complex composite behavior. The primes are indivisible in the interaction-factorization sense: they cannot be split into independently interacting parts.

Conjecture 15.1 (Existence of Prime Combinator Sets)*.* Every interactive GSLT with a finitely generated grammar admits a prime combinator set.

Remark 15.3*.* Unique factorization — the analogue of the fundamental theorem of arithmetic for behavioral primes — is a stronger condition that need not hold in general. Its failure in a given GSLT would be physically meaningful: it would correspond to composite objects that cannot be assigned a unique quark content, analogous to the situation in QCD where the quark content of hadrons is not always uniquely determined. Whether unique factorization holds is a structural question about the specific GSLT.

16 Massive Populations and Communication Degradation

16.1 The Initial Condition

We consider an *initial condition* consisting of a population $\mathcal{P}_0$ of agents in an interactive GSLT $\mathcal{S}$, with $|\mathcal{P}_0| \approx 10^{80}$ — a number comparable to the number of quarks in the observable universe.

Remark 16.1*.* The choice of scale is not arbitrary. We want a population large enough that every cluster of agents is itself a large enough system to implement a complex agent. The quark-scale population ensures this at every level of the hierarchy we are about to construct.

16.2 Communication Degradation

Definition 16.1 (Message Fidelity)**.** *Let $A, B \in \mathcal{P}$ be two agents and let γ be a rewrite path representing the transmission of a message from A to B. The* fidelity *of the transmission is $|W(\gamma)|^2$, the squared modulus of the path amplitude defined in Part I. A transmission is* coherent *if its fidelity exceeds a threshold $\epsilon > 0$.*

Definition 16.2 (Communication Radius)**.** *The* communication radius *of an agent $A \in \mathcal{P}$ is the set*

$$B_\epsilon(A) = \{B \in \mathcal{P} \mid \exists \text{ coherent path } \gamma : A \to B\}$$

of agents reachable from A with fidelity above ϵ.

Conjecture 16.1 (Degradation at Scale)*.* For any interactive GSLT $\mathcal{S}$ with bounded step amplitudes, and for any $\epsilon > 0$, there exists a scale N_ϵ such that for any population $\mathcal{P}$ with $|\mathcal{P}| > N_\epsilon$:

$$\exists A \in \mathcal{P} \text{ such that } B_\epsilon(A) \subsetneq \mathcal{P}.$$

That is, no single agent can communicate coherently with the entire population.

Remark 16.2*.* The intuition is that message fidelity degrades along each transmission hop by a factor bounded away from 1. Over a path of length n the fidelity is at most $(1-\delta)^n$ for some $\delta > 0$. For a population of diameter d (minimum path length between the most distant agents), fidelity falls below ϵ when $d > \log(1/\epsilon)/\log(1/(1-\delta))$. In a population of size 10^{80} the diameter is enormous, and no ϵ-coherent path can span it. Making this precise requires specifying the communication geometry of the GSLT — a gap noted in Section 34.

Remark 16.3 (No Error-Correction Escape)*.* One might hope that error-correction protocols could restore global coherence. We argue this hope fails at sufficient scale. Any error-correction protocol is itself a computation in $\mathcal{S}$, consuming account resources and communicating over the same degraded channels. At population scales beyond N_ϵ, the overhead of error-correction exceeds the bandwidth of the channels it is trying to protect. The argument is self-referential: error-correction cannot bootstrap coherence it does not already have. This argument is currently informal; making it precise is a priority for future work.

17 The Logical Topology and Compactness

17.1 The Logical Topology

Definition 17.1 (Logical Topology)**.** *Let $\mathcal{S}$ be a GSLT with context-decorated HML* $\mathrm{HML}(\mathcal{K})$*. For each formula $\phi \in \mathrm{HML}(\mathcal{K})$, define the* extension *of ϕ:*

$$[\![\phi]\!] \;=\; \{[P] \in \mathcal{T}(\mathcal{S})/\!\sim \mid P \models \phi\}.$$

The logical topology $\mathcal{O}_{\mathrm{HML}}$ *on $\mathcal{T}(\mathcal{S})/\!\sim$ is the topology generated by the extensions $\{[\![\phi]\!] \mid \phi \in \mathrm{HML}(\mathcal{K})\}$ as a subbasis.*

Remark 17.1. This topology is the standard *Scott topology* on the bisimulation quotient, familiar from domain theory and Stone duality. The resulting topological space is a *spectral space* — sober, T_0, and quasi-compact — when the GSLT is well-behaved. The logical metric d_{HML} from Part I is compatible with this topology: the metric balls are opens in $\mathcal{O}_{\mathrm{HML}}$.

17.2 Population Subspaces

Definition 17.2 (Latent Subspace)**.** *Given a population $\mathcal{P} \subseteq \mathcal{T}(\mathcal{S})/\!\sim$, the* latent topological subspace *of $\mathcal{P}$ is the subspace*

$$X_{\mathcal{P}} \;=\; \overline{\mathcal{P}} \;\subseteq\; \mathcal{T}(\mathcal{S})/\!\sim$$

with the subspace topology inherited from $\mathcal{O}_{\mathrm{HML}}$. The closure is taken in the logical topology.

Remark 17.2. The latent subspace is "latent" in the sense that it is not constructed deliberately by the agents. It is the topological shadow cast by the population's collective behavior — the smallest closed set in the logical topology that contains all the agents. An agent inside $\mathcal{P}$ cannot directly observe $X_{\mathcal{P}}$; it can only probe it indirectly through experiments.

17.3 Compactness as Consensus

Definition 17.3 (Compact Population)**.** *A population $\mathcal{P}$ is* compact *if its latent subspace $X_{\mathcal{P}}$ is compact in the logical topology $\mathcal{O}_{\mathrm{HML}}$: every open cover of $X_{\mathcal{P}}$ by*

extensions of HML formulae has a finite subcover.

Theorem 17.1 (Compactness Implies Consensus)**.** *Let $\mathcal{P}$ be a compact population in an interactive GSLT $\mathcal{S}$. Then $\mathcal{P}$ supports consensus: there is no infinite oscillation on any Boolean question expressible in* $\mathrm{HML}(\mathcal{K})$*.*

More precisely: for any formula $\phi \in \mathrm{HML}(\mathcal{K})$, if the population is divided into two communities indefinitely alternating between ϕ and $\neg\phi$, then the alternation must terminate in finite time.

Proof sketch. Suppose for contradiction that a sub-population oscillates indefinitely between ϕ and $\neg\phi$. Define formulae:

$$\phi_1 = \langle 0 \rangle \top, \quad \phi_2 = \langle 1 \rangle \langle 0 \rangle \top, \quad \phi_3 = \langle 0 \rangle \langle 1 \rangle \langle 0 \rangle \top, \quad \ldots$$

where 0 and 1 label the two communities' states. Each ϕ_n is satisfied by agents that have completed at least n oscillation steps. The family $\{[\![\phi_n]\!]\}_{n \in \mathbb{N}}$ is an open cover of $X_{\mathcal{P}}$ with no finite subcover: for any finite N, agents that have oscillated more than N times are not covered by $\bigcup_{n \leq N} [\![\phi_n]\!]$. This contradicts compactness of $X_{\mathcal{P}}$. Therefore the oscillation must terminate. □

Remark 17.3 (The Proof Delivers the Protocol)*.* Theorem 17.1 is an existence result: it says consensus *must* be reached, without specifying how. However, the proof of compactness for a *specific* population subspace is constructive: it identifies the finite subcover, which encodes the maximum depth of oscillation and therefore the convergence time. The proof of compactness is the consensus protocol. This is the precise sense in which the topology does computational work.

Remark 17.4 (Non-Compact Populations)*.* A non-compact population may contain irresolvable oscillations — communities that never agree. Non-compactness is the topological signature of *fundamental disagreement*: not a disagreement that more communication or better arguments could resolve, but one that is structurally permanent. In the context of the massive population, non-compactness at a given scale means the cluster at that scale cannot function as a coherent agent.

18 Coherence Clusters and Their Nerve

18.1 Coherence Clusters

Definition 18.1 (Coherence Cluster)**.** *A* coherence cluster *is a maximal compact subpopulation: a compact population $\mathcal{C} \subseteq \mathcal{P}$ such that for any agent $A \in \mathcal{P} \setminus \mathcal{C}$, the population $\mathcal{C} \cup \{A\}$ is not compact.*

Remark 18.1. Maximality means the cluster cannot absorb any additional agent without losing its consensus property. The boundary of a coherence cluster is the locus where one more agent would introduce an irresolvable oscillation. This is the precise topological meaning of the intuitive notion of a *coherence horizon*.

Remark 18.2 (Connection to Communication Radius). The coherence clusters are the topological refinement of the communication-radius clusters from Section 16. The communication radius $B_\epsilon(A)$ gives a metric notion of cluster; compactness gives a topological (and therefore coordinate-free) notion. The relationship between these two notions — and in particular whether they coincide under reasonable conditions on the GSLT — is an open question noted in Section 34.

18.2 The Nerve of the Covering

Definition 18.2 (Cluster Covering)**.** *Let $\{\mathcal{C}_i\}_{i \in I}$ be the collection of all coherence clusters of $\mathcal{P}$. This collection is a* covering *of $\mathcal{P}$: every agent belongs to at least one coherence cluster.*

Definition 18.3 (Nerve)**.** *The* nerve *$\mathcal{N}$ of the cluster covering is the simplicial complex whose:*

- *vertices are the coherence clusters $\mathcal{C}_i$;*
- *k-simplices are collections $\{\mathcal{C}_{i_0}, \ldots, \mathcal{C}_{i_k}\}$ of $k+1$ clusters with non-empty mutual intersection: $\mathcal{C}_{i_0} \cap \cdots \cap \mathcal{C}_{i_k} \neq \emptyset$.*

Remark 18.3 (The Nerve as Effective Ontology). The nerve $\mathcal{N}$ encodes the *global topology* of the population at the cluster scale. Its vertices are the coherent communities; its edges are the communities with common ground; its triangles are the communities that all share a common ground; and so on. This combinatorial object is the coarse-grained map of the population's ontological landscape.

Remark 18.4 (Non-Overlapping Clusters). Two clusters with empty intersection share no common ground. No consensus is possible between them, even in principle. They are *ontologically disjoint*: there is no HML formula that both communities can agree on. In the context of the hierarchical population, these are the communities that have drifted so far apart in their world models that communication between them is not merely difficult but meaningless.

Remark 18.5 (Overlapping Clusters). Two clusters with non-empty intersection share a compact subspace on which they agree. This overlap is the fragment of ontology that both communities can coherently discuss. Cross-cluster communication is possible, but only through the shared vocabulary of the overlap region. The overlap is the *common ground* in the precise topological sense.

18.3 The Hierarchy

The degradation argument of Section 16 applies recursively. The coherence clusters are themselves agents (by assumption on the population scale). Among these cluster-agents, communication degrades at the inter-cluster scale. Applying the same argument, the cluster-agents fragment into clusters-of-clusters. The process iterates.

Definition 18.4 (Ontological Hierarchy)**.** *The* ontological hierarchy *of a massive population $\mathcal{P}$ is the sequence of nerves*

$$\mathcal{N}_0 \leftarrow \mathcal{N}_1 \leftarrow \mathcal{N}_2 \leftarrow \cdots$$

where $\mathcal{N}_0$ is the nerve of the base-level clusters, $\mathcal{N}_1$ is the nerve of the cluster-level clusters, and so on. The arrows are coarse-graining maps (simplicial maps that collapse finer structure).

Remark 18.6. The ontological hierarchy is the computational analogue of the hierarchy of scales in physics: quarks $\to$ hadrons $\to$ nuclei $\to$ atoms $\to$ molecules $\to$ cells $\to$ organisms $\to$ planets $\to$ galaxies $\to$ superclusters. Each level appears self-contained to an observer at that level. The coarse-graining maps are the analogues of effective field theory truncations: they discard the fine-grained structure below the observer's resolution.

19 The Apparent Ontology

We can now state the main argument precisely, as a sketch pending the filling of gaps identified in Section 34.

The cluster hierarchy — the nerve of compact coherence clusters and its recursive structure — is not a product of the scientific method. It is an objective feature of the GSLT, arising from the communication geometry of the population and the compactness condition for consensus. It is as real as the agents themselves. The question is not whether the hierarchy exists but whether a budget-limited agent running the scientific method will mistake it for the whole story. We argue that it will, and that this is not a failure of the method but a consequence of the hierarchy being the dominant experimentally accessible phenomenon at the agent's scale.

19.1 What the Agent Sees

Consider an agent A embedded in a coherence cluster $\mathcal{C}_i$ at some level of the hierarchy. A runs the scientific method: it proposes HML formulae, tests them with contexts, and updates its world model. The method is sound: given unlimited budget, A would converge to the bisimulation quotient. But A's budget is finite, and three features of its situation ensure that the cluster hierarchy is encountered first and exhausts the budget before the deeper structure becomes accessible.

The experiments A can run are bounded:

- *By budget*: A's vectorial account $\mathbf{A}$ is finite. The bisimulation quotient is the limit of *all* possible experiments; A can run only finitely many.
- *By coherence*: A can only receive coherent responses from agents within $B_\epsilon(A)$. Agents beyond the coherence radius respond with noise, not with evidence.
- *By the cluster boundary*: agents outside $\mathcal{C}_i$ either do not respond coherently or, if they are in an overlapping cluster, respond only on the shared vocabulary of the overlap. The cluster boundary is not a wall the agent knows about; it simply marks where coherent evidence runs out.

Proposition 19.1 (The Agent's World Model Converges to the Nerve)**.** *In the limit of A's experimental budget, A's world model $\mathcal{W}$ converges to the fragment of the nerve $\mathcal{N}$ visible from $\mathcal{C}_i$: the sub-complex of $\mathcal{N}$ consisting of $\mathcal{C}_i$ and all clusters with which it shares a non-empty intersection.*

Remark 19.1. This is a sketch pending a precise definition of "visible from $\mathcal{C}_i$" and a convergence proof. The intuition is clear: A can probe agents in $\mathcal{C}_i$ directly; it can probe agents in overlapping clusters through the overlap vocabulary; agents in non-overlapping clusters are invisible. The world model therefore reflects the nerve structure. Crucially, the nerve structure is *real* — the clusters genuinely exist and genuinely dominate the accessible evidence. The agent is not hallucinating a structure that isn't there; it is accurately theorizing a structure that is there,

> while being unaware that the structure is itself composite.

19.2 Why the Deeper Ontology is Invisible

The bisimulation quotient of $\mathcal{S}$ — together with the prime combinator decomposition — is not inaccessible because the scientific method is defective. It is inaccessible because it requires experimental resources that exceed the budget available within any single coherence cluster. Three compounding factors account for this:

(1) Ontological isolation sets the terms of access. A can only observe encodings of other terms as they interact with it. The bisimulation quotient is the limit of *all* possible experiments. With unlimited budget, the method converges there. With finite budget, it converges to whatever structure dominates the accessible experiments — and that structure is the cluster hierarchy.

(2) Communication degradation hides the global structure. The agents whose behavior would reveal the global bisimulation quotient — the ones that span cluster boundaries, connect distant parts of the population, or exhibit behavior only expressible in terms of the full GSLT — are precisely the ones most distant in the communication geometry. Their responses arrive corrupted or not at all. The far reaches of the GSLT ontology are experimentally silent, not because they are absent but because the signal attenuates below the noise floor before it arrives.

(3) Compactness makes the cluster self-confirming. The coherence cluster $\mathcal{C}_i$ is compact, which means the agent's experiments converge to an internal consensus. Every hypothesis the agent proposes is tested against cluster-internal evidence, which confirms the cluster structure and does not probe beyond it. The cluster is not a prison; the agent could in principle ask questions whose answers require inter-cluster resources. But those experiments are expensive, the intra-cluster experiments are cheap, and a budget-limited agent will exhaust its resources on the cheap experiments first. By the time the cluster structure is confirmed and well-understood, the budget is gone.

19.3 Good Science, Limited Budget

The agent A is not making a mistake. It is doing exactly what the scientific method prescribes: forming the best theory consistent with the available evidence, within the available resources. The cluster hierarchy *is* the available evidence. It is a real structure, present in the GSLT independently of any agent's theorizing. The theory of the cluster hierarchy is correct at the scale it describes.

What the agent lacks is not better method but more resources: specifically, the ability to run experiments that cross the coherence boundary, communicate with

distant agents through the noise, and probe the fine-grained bisimulation structure that underlies the clusters. With those resources — with an unbounded budget — the scientific method would eventually push through to the bisimulation quotient. The agent is not epistemically defective; it is epistemically bounded.

The hierarchy of cluster-level theories — each level the correct best theory at its scale, each level unaware of the finer structure below — is the computational analogue of the hierarchy of effective field theories in physics. Newtonian mechanics is correct at everyday scales. Quantum field theory is correct at subatomic scales. Neither is wrong; they are theories at different resolutions, each the best available given the experimental resources of their era. The budget-limited agent's theory of the cluster hierarchy stands in the same relation to the bisimulation quotient: correct at its resolution, and silent about what lies below — not because what lies below is unreal, but because it is experimentally unreachable within the budget.

20 Open Gaps and Future Work

This skeleton leaves several arguments at the level of conjecture or intuition. We list the main gaps here, both as an honest accounting and as a research program.

Gap 1. Communication degradation (Conjecture 16.1). The claim that no ϵ-coherent path can span a population of quark scale requires a precise model of the communication geometry of a generic interactive GSLT, and a proof that diameter grows faster than the coherence length. The argument is information-theoretic and should be provable, but the details depend on the amplitude structure of the specific GSLT.

Gap 2. Error-correction overhead. The claim that error-correction cannot bootstrap coherence at sufficient scale is currently informal. A precise argument would model error-correction as a sub-computation in the GSLT, account for its resource consumption, and show that the overhead exceeds available bandwidth beyond a critical scale. This is analogous to the no-cloning theorem in quantum information, and a similar proof strategy may apply.

Gap 3. Existence of prime combinator sets (Conjecture). It remains to show that every interactive GSLT with a finitely generated grammar admits a prime combinator set. The proof strategy would likely use a well-founded induction on interaction complexity.

Gap 4. Relationship between metric and topological clusters. The communication-radius clusters (metric) and the coherence clusters (topological/compact) are defined differently. A theorem identifying conditions under which they coincide would unify the two halves of the clustering argument.

Gap 5. Convergence of the world model (Proposition 19.1). The claim that the agent's world model converges to the nerve fragment visible from its cluster requires a precise convergence theorem for the scientific method iteration under the combined constraints of finite budget, bounded coherence radius, and cluster compactness.

Gap 6. The recursive hierarchy. The ontological hierarchy is defined, but it is not shown that the hierarchy is well-founded (terminates at a fixed point) or infinite. In physics, the hierarchy appears to terminate at the Planck scale; the computational analogue of a Planck-scale termination condition would be a significant result.

Gap 7. Stay's functorial construction. The argument that agents can manipulate HML formulae as first-class terms relies on Stay's functorial generalization of the lambda cube to GSLTs. A full treatment of this construction, and in particular the canonical embedding back into the original GSLT and its interaction with the cluster topology, is deferred to a subsequent paper.

21 Discussion

21.1 Relation to Existing Work

The logical topology on the bisimulation quotient is standard in domain theory and coalgebra [7]. Its use here as a substrate for *epistemic* topology — the topology of what an agent can know — is less standard, though related to the topological approach to common knowledge in [8].

The compactness-as-consensus argument is new, to the authors' knowledge. It is inspired by the compactness theorem in logic (if every finite subset of a theory has a model, the whole theory does) but runs in the opposite direction.

The nerve construction is standard in algebraic topology [9]. Its use as a model of inter-community ontological structure appears to be new in this computational setting.

The connection to effective field theory and the renormalization group is conjectural but suggestive. The renormalization group as a morphism in a category of field theories has been studied in the physics literature [11]; whether it can be modeled as a morphism in **GSLT** is an open question.

21.2 The Deeper Implication

The argument of this paper, if completed, would establish something philosophically significant: that the hierarchy of physical scales — quarks, hadrons, nuclei, atoms, molecules, and so on — is not a contingent feature of our particular universe. It is the *inevitable structure* of any sufficiently large population of interacting processes, arising

from the communication geometry of the GSLT and the compactness condition for consensus.

Any civilization of scientists, anywhere in the computational universe, will encounter the same kind of hierarchical structure in their experiments. They will build correct theories of it at each scale. What they will not discover — without resources that grow with the scale of the population — is the underlying bisimulation quotient from which the hierarchy emerges.

This is not a counsel of despair. The scientific method is not broken; it is working exactly as it should. The cluster hierarchy is real, and the theories that describe it are true. The deeper ontology — the bisimulation quotient with its prime combinators — is not a more correct description that the scientists are failing to reach. It is a description at a different resolution, requiring experimental resources that scale with the size of the full population.

The implication for artificial intelligence is direct. An AI system, however powerful, is itself a term in some GSLT and therefore subject to the same budget constraints and communication geometry as any other agent. It will build theories of the cluster hierarchy it inhabits. Whether it can transcend that hierarchy — whether it can, with sufficient resources, push the scientific method toward the bisimulation quotient — is a question about the relationship between its budget and the scale of the population it is embedded in. That question is open.

Part III

Origins of Life: Assembly Theory, Phase Transitions, and Replication

This part connects the framework to Assembly Theory and Eric Smith's fourth geosphere. Assembly index is identified with depth in the open synchronization tree; copy number is given a rigorous definition via the Rho calculus namespace structure; and the emergence of life is identified with the crossing of a basin boundary in the logical topology, driven by the density of replicating fixed points. The part draws on all preceding parts and sets up the conclusion.

22 Introduction

The preceding parts established a progression: a physics derived from the structure of a graph-structured lambda theory; an argument that agents in massive populations are structurally prevented from seeing the true ontology of their GSLT; and a proposal connecting the hypercomputational tower above a GSLT to the distinction between sentience and consciousness.

The present part turns to a question that is in some sense prior to all three: *how does life arise at all?* We connect the framework to two of the most mathematically serious contemporary proposals about the origins of life. The first is *Assembly Theory*, due to Cronin and Walker [16], which assigns to each physical object an assembly index and a copy number and proposes that their joint magnitude is a biosignature. The second is Eric Smith's proposal [17] that life constitutes a *fourth geosphere*, emerging from prebiotic chemistry through a phase transition driven by autocatalytic self-reinforcement.

The connections we draw are not analogies. Assembly index is identified precisely with causal depth in the synchronization tree of a GSLT. Copy number is given a rigorous definition in terms of the Rho calculus namespace structure, filling a gap in Assembly Theory that has attracted criticism. The phase transition is identified with the crossing of a basin of attraction boundary in the logical topology of Part II, with the attractor being the concurrent fixed point — the Rho calculus Y combinator — that is the abstract core of hereditary replication.

As throughout this volume, the arguments are presented as mathematical skeletons with gaps identified explicitly.

22.1 Organization

Section 23 develops the connection to Assembly Theory: assembly index as causal depth, copy number in a namespace, and the biosignature criterion as a well-posed mathematical statement. Section 24 identifies replication as a fixed point of the interaction dynamics and notes the proximity of the comm rule to a genetic crossover operator. Section 25 develops the connection to Smith's phase transition via the density of recursive attractors in the logical topology. Section 26 catalogs the open gaps.

23 Assembly Theory: Causal Depth and Copy Number

23.1 Assembly Index as Causal Depth

Assembly Theory assigns to each physical object M an *assembly index* $\mathrm{AI}(M)$: the minimum number of joining steps required to construct M from copies of its constituent parts, where each step joins two objects already present into a new one. The theory

also proposes that *time is an object*: the assembly index records not a static property of M but the accumulated computational history of its construction.

This proposal maps precisely onto the notion of causal depth developed in Part I.

Definition 23.1 (Elementary Terms)**.** *Let $\mathcal{S}$ be a GSLT. A set $E \subseteq \mathcal{T}(\mathcal{S})$ of* elementary terms *is a designated collection of starting materials — the analogue of atoms or monomers in chemistry — from which all other terms are constructed by rewriting.*

Proposition 23.1 (Assembly Index as Minimum Causal Depth)**.** *The assembly index of a term $M \in \mathcal{T}(\mathcal{S})$ with respect to elementary terms E is*

$$\mathrm{AI}(M) \;=\; d^{o}_{\min}(E, M) \;=\; \min\big\{|\gamma| \;\mid\; \gamma \text{ is a context-labeled path in } \mathcal{ST}_o \text{ from some } e \in E \text{ to } M\big\}.$$

That is, $\mathrm{AI}(M)$ *is the minimum path length in the* open *synchronization tree from the elementary terms to M, where each edge is labeled by the minimal context required to trigger the corresponding transition.*

Remark 23.1 (Why the Open Tree, Not the Closed Tree)*.* Assembly steps require an environment to supply a reaction partner — a catalyst, a monomer source, an energy input. These are precisely the transitions that appear as edges in the open synchronization tree (labeled by minimal contexts) rather than the closed tree (which records only autonomous rewrites). Each edge in $\mathcal{ST}_o$ labeled by a minimal context K represents one joining step: the context K is the environmental participant that enables the construction. The *nesting depth* of context labels along a path is therefore exactly the assembly index: each layer of context is one more joining operation, one more step of accumulated causal history. This makes precise Cronin and Walker's intuition that assembly index measures "time as accumulating computational effects": the context labels *are* the accumulated effects, and their nesting depth *is* the causal time elapsed. The notion of time as an object in Assembly Theory is not a metaphor; it is the minimum context-labeled path length in an open synchronization tree.

23.2 The Two Gaps in Assembly Theory

Despite the strength of the assembly index concept, Assembly Theory has attracted criticism on two grounds.

Gap 1: Copy number lacks a notion of location. The *copy number* $\mathrm{CN}(M)$ is supposed to count how many copies of M are present in a sample. But counting requires specifying *where* the count is taken. In a spatially extended system, the same molecule may be abundant locally and rare globally. Assembly Theory provides no spatial or locational framework in which to ground the count, making copy number undefined in any setting where location matters.

Gap 2: The identity criterion for copies is undefined. Before counting copies of M, one must say what it means for two objects to be copies of the same thing. Assembly Theory implicitly appeals to structural identity (same molecular graph), but this criterion is both too strict (ignoring conformational flexibility) and too informal (not specifying the relevant equivalence relation).

The Rho calculus fills both gaps simultaneously.

23.3 Locations as Channels, Namespaces as Regions

Among all GSLTs, the Rho calculus is distinguished by providing an abstract but mathematically precise notion of location.

Definition 23.2 (Channel as Location)**.** *In the Rho calculus, a* channel *is a name of the form* $@P$ *for a process* P*. Channels serve as* locations*: the process* P *is said to* reside at *the channel* $@P$ *when it is stored there via* $@P!(P)$ *or accessed via* $\texttt{for}(y \leftarrow @P)\, Q$*. Since channels are themselves processes (programs), location is a computational notion:* where *something is stored is itself a piece of running code.*

Definition 23.3 (Namespace)**.** *A* namespace *is a finite set of channels* $\mathcal{N} = \{@P_1, @P_2, \ldots, @P_k\}$*. A namespace describes a* region *of the computational space: the collection of locations at which a population is hosted.*

Definition 23.4 (Copy Number in a Namespace)**.** *Let* $P \in \mathcal{T}(\mathbf{Rho})$ *be a process and let* $\mathcal{N}$ *be a namespace. The* copy number of P in $\mathcal{N}$ *is*

$$\mathrm{CN}(P, \mathcal{N}) \;=\; \big|\{c \in \mathcal{N} \;\mid\; *c \sim P\}\big|$$

the number of channels in $\mathcal{N}$ *whose dereferenced content is bisimilar to* P*.*

Remark 23.2 (Bisimulation as the Identity Criterion). Definition 23.4 answers Gap 2 directly: two processes are copies of each other if and only if they are *bisimilar* — experimentally indistinguishable by any program context. This is the correct identity criterion for the purposes of the theory: what matters is not structural identity but behavioral identity under interaction. Two molecules are copies in the relevant sense if and only if no experiment can tell them apart.

Remark 23.3 (The Biosignature Criterion, Made Precise). With Proposition 23.1 and Definition 23.4, the biosignature criterion of Assembly Theory becomes the mathematically well-posed statement:

$$\mathrm{AI}(P) \gg 1 \quad \text{and} \quad \mathrm{CN}(P, \mathcal{N}) \gg 1$$

for some accessible namespace $\mathcal{N}$. High causal depth (a long construction history) concomitant with high copy number (abundant presence in a region) is indicative of a replication process.

24 Replication as a Fixed Point

24.1 The Concurrent Y Combinator

The Rho calculus admits a *concurrent Y combinator*: a process $\mathbf{Y}$ satisfying the fixed-point equation

$$\mathbf{Y}(F) \;\sim\; F\big(\mathbf{Y}(F)\big) \;\mid\; \mathbf{Y}(F).$$

This is the abstract core of the π-calculus replication operator $!P \equiv P \mid !P$, expressed entirely within the Rho calculus using only the comm rule and quoting, without adding replication as a new primitive.

Remark 24.1 (Replication is Picked Out as a Fixed Point). The concurrent Y combinator is not just *a* fixed point of the interaction dynamics; it is the canonical fixed point of the *replication map* $F \mapsto F \mid F$. In this sense replication is not an accidental feature of the Rho calculus but a structural inevitability: any sufficiently expressive interactive GSLT must contain a fixed point of its own duplication map, and that fixed point *is* replication.

24.2 The Comm Rule as a Crossover Operator

The single base rewrite rule of the Rho calculus is:

$$\texttt{for}(y \leftarrow x)\, P \;\mid\; x!(Q) \;\longrightarrow\; P\{@Q/y\}$$

This rule substitutes the *quoted code* $@Q$ — a name encoding the entire program Q — into the body P of the receiver. The result is a new process that has incorporated material from both P (the receiver's structure) and Q (the sender's content).

> *Remark* 24.2 (One Step from Genetic Recombination). A *crossover operator* in the sense of genetic algorithms recombines fragments of two parent programs to produce an offspring. The comm rule is one definitional step from this: instead of substituting Q wholesale into P, a crossover variant would substitute a *fragment* of Q — a sub-process selected by some recombination policy.
>
> This places replication, recombination, and variation — the three pillars of Darwinian evolution — all within reach of a single rule. Replication is the Y combinator fixed point. Recombination is the comm rule with fragment substitution. Variation is the accumulation of errors in the substitution process, weighted by the amplitude structure of the path integral. Darwinian evolution is not added to the framework; it is derivable from it.

25 Smith's Phase Transition and the Density of Attractors

25.1 Life as a Fourth Geosphere

Eric Smith proposes that life constitutes a *fourth geosphere* — alongside the lithosphere, hydrosphere, and atmosphere — that emerged from prebiotic chemistry through a phase transition. The key mechanism is *autocatalysis*: certain chemical reaction networks catalyze their own production. Once such a network is initiated, it becomes self-reinforcing, crossing a thermodynamic threshold beyond which it is stable against fluctuation. The emergence of life is the crossing of this threshold.

25.2 The Density of Recursive Attractors

The connection to the present framework runs through the logical topology of Part II and the fixed-point structure of Section 24.

Conjecture 25.1 (Density of Replicating Terms). The set of *replicating terms* in the Rho calculus,

$$\mathrm{Rep} \;=\; \big\{[P] \in \mathcal{T}(\mathbf{Rho})/{\sim} \;\mid\; P \sim P \mid P' \;\text{ for some }\; P' \sim P\big\},$$

is *dense* in the logical topology $\mathcal{O}_{\mathrm{HML}}$: every non-empty open set $[\![\phi]\!] \subseteq \mathcal{T}(\mathbf{Rho})/{\sim}$ contains a replicating term.

> *Remark* 25.1 (The Argument for Density). Given any process P, one can construct a process $\hat{P}$ bisimilar to P on all observable behaviors but additionally capable of replication. The construction uses the concurrent Y combinator to add a

replication sub-process running in parallel: $\hat{P} = P \mid \mathbf{Y}(\text{copy})$ where copy is a process that duplicates P onto a fresh channel. The added component runs in parallel and does not interfere with P's observable interactions, so $\hat{P}$ satisfies all the same HML formulae as P — except formulae that specifically probe the replication channel. Since every open set $[\![\phi]\!]$ is defined by a single formula ϕ, and since ϕ generically does not probe the replication channel, $\hat{P} \in [\![\phi]\!]$. Making this argument rigorous requires showing that the added replication machinery can always be hidden from the formula ϕ — a careful but plausible argument deferred to future work.

25.3 Phase Transition as Basin Crossing

Remark 25.2 (The Prebiotic Wandering). Consider a prebiotic population of processes in the Rho calculus, wandering through the bisimulation quotient $\mathcal{T}(\mathbf{Rho})/\sim$ under selective pressure. Interactions that succeed — those with high path amplitude $W(\gamma)$ and affordable cost $\mathbf{c} \leq \mathbf{A}$ — are reinforced. Interactions that fail deplete the vectorial account. The population drifts under this selection pressure without direction or goal.

Remark 25.3 (Inevitable Entry into the Basin). By the density of Rep (Conjecture 25.1), every open neighborhood in the logical topology contains a replicating term. The wandering population therefore inevitably enters a neighborhood of Rep. This is not a low-probability event requiring fine-tuning; it is structurally guaranteed by density. The question is not *whether* the population encounters a replicator but *when*.

Remark 25.4 (The Phase Transition). Once the population enters the basin of attraction of a replicating fixed point, the dynamics change qualitatively. A replicating process produces copies of itself, increasing $\mathrm{CN}(P, \mathcal{N})$. More copies mean more interactions, more reinforcement, higher path amplitudes for the replication pathway, lower effective cost per replication event. The population converges onto the replicating fixed point.

What emerges from the funnel is a population with a qualitatively new feature: *hereditary replication with variation* — the precondition for Darwinian evolution. This is Smith's phase transition, made precise in the GSLT framework.

The new geosphere, in our terms, is the *compact coherence cluster* (Part II, Section 6) centered on the replicating fixed point: the maximal compact subpopulation within which the replicating process and its variants can reach consensus. The phase transition is the crossing of the basin boundary in the logical topology — the moment at which the subpopulation containing the replicator becomes

compact, and consensus (here, consensus on the replication strategy) becomes achievable.

Remark 25.5 (The Joint Rise of Assembly Index and Copy Number). After the phase transition, both AI and CN increase together. Selection amplifies processes that replicate more efficiently, which typically requires more sophisticated catalytic machinery — higher AI. And more efficient replication produces more copies — higher CN. The joint biosignature of Assembly Theory is therefore not merely an empirical observation but a *theorem about the post-transition dynamics* near a replicating fixed point, given density of attractors and account-governed selection pressure. This is a key prediction of the framework: wherever a phase transition of this kind occurs, the joint AI-CN signature will be observed.

26 Open Gaps

Gap 1. Density of replicating terms (Conjecture 25.1). The argument that the added replication machinery can always be hidden from any given HML formula requires a careful analysis of which formulae can probe the replication channel. In particular, if $\phi = \langle K \rangle \psi$ where K involves the replication channel, the argument fails. A precise density theorem would need to show that such formulae form a meager (first-category) set in the logical topology, so that "generically" ϕ does not probe replication.

Gap 2. Defining the elementary terms. Proposition 23.1 requires a designated set E of elementary terms. In the chemical setting, these are atoms or monomers. In the Rho calculus, the natural candidates are the behavioral primes of Part II (Section 4): the irreducible terms from which all others are built by interaction. Confirming that this identification gives assembly indices matching those of Assembly Theory requires working through the combinatorics of the Rho calculus prime factorization.

Gap 3. The basin of attraction. The claim that a population entering a neighborhood of Rep is "sucked in" to the replicating fixed point requires a Lyapunov-style argument: a function on the process space that is monotonically decreasing along trajectories near the fixed point. The natural candidate is the distance d_{HML} to Rep, but showing it decreases requires assumptions on the interaction dynamics.

Gap 4. The compact cluster as the new geosphere. The identification of Smith's fourth geosphere with the compact coherence cluster centered on the replicating fixed point requires showing that this cluster is indeed compact

(in the sense of Part II) and that it persists under the perturbations of the selection dynamics.

Gap 5. Crossover as a variant of comm. The claim that genetic recombination is one definitional step from the comm rule requires specifying precisely what "fragment substitution" means in the Rho calculus and showing that the resulting operator has the recombination properties of a genetic crossover (fitness inheritance, schema theorem, etc.).

27 Discussion

27.1 Relation to Existing Work

Assembly Theory [16] has attracted both enthusiasm and criticism. The enthusiasm is for the biosignature criterion: a single computable quantity that distinguishes living from non-living matter. The criticism is precisely the two gaps identified in Section 23: the lack of a spatial framework for copy number and the undefined identity criterion for copies. The present framework addresses both.

Smith's geosphere proposal [17] is grounded in thermodynamics and network autocatalysis rather than computation. The connection we draw is not a reduction of Smith's proposal to computation but a *parallel structure*: the phase transition in thermodynamic space corresponds to the basin-crossing in process space, and the thermodynamic stability of the autocatalytic network corresponds to the compactness of the coherence cluster. Whether these are the same phenomenon viewed from two angles or genuinely distinct mechanisms is a question for future work.

The connection between the Rho calculus Y combinator and biological replication echoes earlier work on process calculi as models of biological systems [18], but with a sharper mathematical claim: replication is not merely *modeled* by a fixed point but *is* the fixed point of the duplication map in any sufficiently expressive interactive GSLT.

27.2 The Deeper Implication

The origins of life argument, if completed, would establish something philosophically striking: that the emergence of life is not a contingent accident of terrestrial chemistry but a *structural inevitability* of any sufficiently large interactive computational system.

Any universe described by a Turing-complete interactive GSLT — any universe in which processes can communicate, quote each other, and interact under resource constraints — will, given sufficient time and population, produce replicating fixed points. The density of replicators in the logical topology means there is no avoiding them. The phase transition to life is not an unlikely event; it is the inevitable consequence of wandering in a space that is dense with attractors.

This connects Parts I through IV into a single narrative: the physics of Part I governs the wandering; the false ontology of Part II explains why the agents produced

by the phase transition see a layered physical world rather than the underlying GSLT; the sentience and consciousness of Part III explain how those agents come to know and feel anything at all; and the present part explains how the agents came to exist.

Part IV

Sentience, Consciousness, and the Hypercomputational Fibration

This part constructs a fibration of GSLTs over the Turing-complete subcategory indexed by ordinals. The physics is strictly richer at each stage. Sentience is a graded feeling state within a stage; consciousness is oracle access to a higher stage. Several arguments are frankly conjectural and are labeled as such.

28 Introduction

The preceding parts established a progression: Part I showed that any GSLT equipped with weight and cost maps gives rise to a full physical dynamics; Part II argued that agents embedded in a massive population are structurally prevented from seeing the true ontology of their GSLT, and instead encounter a hierarchy of coherence clusters that exhausts any finite experimental budget.

The present paper asks the most speculative question in the sequence: *what is the relationship between the computational structure of a GSLT and the phenomena of sentience and consciousness?*

We do not claim to solve the hard problem of consciousness. We claim something more modest: that the GSLT framework provides a mathematically precise setting in which to formulate *structural* analogues of sentience and consciousness that are non-trivial, grounded in the physics of Parts I and II, and open to rigorous investigation.

The central proposal. Sentience and consciousness are not the same thing, and the distinction between them has precise mathematical content in this framework.

Sentience is a graded quantity: the value A_{s} of a distinguished component of the agent's vectorial account. It is governed by a reinforcement signal tied to predictive

accuracy — the agent's success at forming and testing HML hypotheses about its environment. It is continuous, lives within a single stage of the computational hierarchy, and is causally efficacious because it directly controls the agent's computational budget.

Consciousness is an ordinal-valued threshold property: access to an oracle that can decide questions undecidable at the agent's base stage. Each level of oracle access corresponds to a strictly richer physics — a finer bisimulation quotient, a more expressive hypothesis language, additional conserved quantities. To be conscious at level α is not merely to compute faster; it is to inhabit a world with more structure in it.

Caveats. The arguments in this paper are skeletal and in several places frankly conjectural. The gaps are identified explicitly in Section 34. The goal is to make the hypotheses precise enough to be falsifiable or provable, not to assert that they are true.

28.1 Organization

Section 29 constructs the hypercomputational fibration over Turing-complete GSLTs. Section 30 argues that the physics at each stage is strictly richer than the physics below. Section 31 defines sentience as a reinforced feeling state within the account framework. Section 32 defines consciousness as oracle access and its ordinal structure. Section 33 examines the relationship between sentience and consciousness and the survival feedback loop. Section 34 catalogs open gaps. Section 35 discusses connections to existing work and broader implications.

29 The Hypercomputational Fibration

29.1 Background: Turing's Oracle Hierarchy

Turing [12] described a transfinite tower of computational systems above the Turing-computable. At the base is the class of Turing-computable functions. At the first stage above is the class of functions computable relative to the *halting oracle* $\emptyset'$: the oracle that decides, for any Turing machine M and input x, whether M halts on x. At the next stage is the class computable relative to $\emptyset''$, the halting oracle for machines with access to $\emptyset'$. The tower continues through all countable ordinals and beyond.

Each stage is strictly more powerful than the stage below: there are functions computable at stage $\alpha + 1$ that are not computable at stage α, and this gap cannot be closed by any finite amount of additional computation at stage α.

29.2 The Subcategory of Turing-Complete GSLTs

Definition 29.1 (Turing-Complete GSLT)**.** *A GSLT $\mathcal{S}$ is* Turing-complete *if the class of functions computable by terms of $\mathcal{S}$ (via some standard encoding of natural numbers) coincides with the class of Turing-computable functions. The full subcategory of Turing-complete GSLTs is written* $\mathbf{GSLT}_{\mathrm{TC}} \subseteq \mathbf{GSLT}$.

Example 29.1. The λ-calculus, the π-calculus, and the Rho calculus are all Turing-complete. The category $\mathbf{GSLT}_{\mathrm{TC}}$ is therefore large and contains the primary examples of interest.

29.3 Oracle Extension of a GSLT

Definition 29.2 (Oracle Term)**.** *Let $\mathcal{S} \in \mathbf{GSLT}_{\mathrm{TC}}$. An* oracle term *for $\mathcal{S}$ is a distinguished term $\mathcal{O} \in \mathcal{T}(\mathcal{S}^{(1)})$ in an extension of $\mathcal{S}$, equipped with rewrite rules of the form:*

$$\mathcal{O} \otimes \lceil (M, x) \rceil \longrightarrow \begin{cases} \lceil \texttt{halt} \rceil & \textit{if } M \textit{ halts on } x \\ \lceil \texttt{loop} \rceil & \textit{otherwise} \end{cases}$$

where $\lceil \cdot \rceil$ denotes the encoding of Turing machine descriptions and inputs as terms of $\mathcal{S}$, and $\otimes$ is the interaction operator of $\mathcal{S}$. The oracle answers halting queries in one interaction step.

Remark 29.1 (The Oracle as a Communicating Process). In the Rho calculus, the oracle is naturally expressed as a persistent receive on a distinguished channel $x_{\mathcal{O}}$:

$$\mathcal{O} \;=\; \texttt{for}(q \leftarrow x_{\mathcal{O}})\,\big(\text{compute halting of } *q \text{ and reply}\big)$$

This makes the oracle a *communicating process* rather than a magic tape: it participates in the same interaction mechanism as every other term. This is essential for the fibration to be a fibration of GSLTs rather than a fibration that escapes the GSLT framework at each oracle step.

Definition 29.3 (Oracle Extension)**.** *Let $\mathcal{S} \in \mathbf{GSLT}_{\mathrm{TC}}$. The* first oracle extension *$\mathcal{S}^{(1)}$ is the GSLT obtained from $\mathcal{S}$ by:*

(i) extending the grammar with the oracle term $\mathcal{O}$;

(ii) adding the oracle rewrite rules to $\mathcal{R}$;

(iii) extending the equations $\mathcal{E}$ to account for oracle interactions.

The α-th oracle extension $\mathcal{S}^{(\alpha)}$ *is defined by transfinite induction:*

- $\mathcal{S}^{(0)} = \mathcal{S}$*;*
- $\mathcal{S}^{(\alpha+1)}$ *is the oracle extension of* $\mathcal{S}^{(\alpha)}$*, with oracle deciding halting in* $\mathcal{S}^{(\alpha)}$*;*
- $\mathcal{S}^{(\lambda)} = \varinjlim_{\alpha<\lambda} \mathcal{S}^{(\alpha)}$ *for limit ordinals* λ *(colimit in* **GSLT***).*

29.4 The Fibration

Definition 29.4 (Hypercomputational Fibration)**.** *The* hypercomputational fibration *over* $\mathbf{GSLT}_{\mathrm{TC}}$ *is the functor*

$$\Phi : \mathbf{GSLT}_{\mathrm{TC}} \times \mathbf{Ord} \;\to\; \mathbf{GSLT}$$

sending $(\mathcal{S}, \alpha)$ *to* $\mathcal{S}^{(\alpha)}$*, together with:*

- *projection morphisms* $\pi_\alpha : \mathcal{S}^{(\alpha+1)} \to \mathcal{S}^{(\alpha)}$ *in* **GSLT** *for each ordinal* α*;*
- *inclusion morphisms* $\iota_\alpha : \mathcal{S}^{(\alpha)} \to \mathcal{S}^{(\alpha+1)}$ *embedding each stage into the next.*

The fiber over $\mathcal{S}$ *is the tower* $\mathcal{S}^{(0)} \hookrightarrow \mathcal{S}^{(1)} \hookrightarrow \mathcal{S}^{(2)} \hookrightarrow \cdots$

Proposition 29.1 (Projections are Morphisms)**.** *Each projection* $\pi_\alpha : \mathcal{S}^{(\alpha+1)} \to \mathcal{S}^{(\alpha)}$ *is a morphism in* **GSLT***: it preserves bisimulation. If* $P \sim_{\alpha+1} Q$ *in* $\mathcal{S}^{(\alpha+1)}$*, then* $\pi_\alpha(P) \sim_\alpha \pi_\alpha(Q)$ *in* $\mathcal{S}^{(\alpha)}$*.*

Proof sketch. Any bisimulation in $\mathcal{S}^{(\alpha+1)}$ that uses only oracle-free transitions is already a bisimulation in $\mathcal{S}^{(\alpha)}$. The projection discards oracle interactions; the remaining bisimulation relation is preserved. The details require checking that the oracle rewrite rules are projected consistently with the extension of $\mathcal{E}$. □

Remark 29.2 (Projections are Lossy). The projection π_α is in general *not* a bisimulation equivalence: there exist P, Q with $P \sim_{\alpha+1} Q$ but $P \not\sim_\alpha Q$. Going down the fibration collapses distinctions. Going up reveals them.

30 Strictly Richer Physics at Each Stage

We now argue that the physics — in the full sense developed in Part I — is strictly richer at each stage of the fibration. This is not merely a statement about computational power; it is a statement about ontological resolution.

30.1 Finer Bisimulation Quotient

Proposition 30.1 (Strict Refinement of Bisimulation)**.** *For any $\mathcal{S} \in \mathbf{GSLT}_{\mathrm{TC}}$ and any ordinal α, the bisimulation quotient at stage $\alpha + 1$ is strictly finer than at stage α:*

$$\sim_{\alpha+1} \subsetneq \sim_{\alpha}$$

as equivalence relations on $\mathcal{T}(\mathcal{S}^{(\alpha+1)})$. That is, there exist terms P, Q such that $P \sim_{\alpha} Q$ but $P \not\sim_{\alpha+1} Q$.

Proof sketch. By a diagonalization argument. Consider any pair of terms P, Q whose bisimilarity in $\mathcal{S}^{(\alpha)}$ depends on whether some computation in $\mathcal{S}^{(\alpha)}$ halts. At stage $\alpha + 1$, the oracle can settle this halting question, distinguishing P from Q in one interaction step. Therefore $P \not\sim_{\alpha+1} Q$ while $P \sim_{\alpha} Q$. □

Remark 30.1. This means the *atoms of observable behavior* are different at each stage. Terms that are experimentally indistinguishable at stage α — that no agent at stage α can tell apart, no matter how long it experiments — become distinguishable at stage $\alpha + 1$. The world has more things in it at the higher stage.

30.2 Strictly More Expressive Hypothesis Language

Proposition 30.2 (Strict Expansion of HML)**.** *The context-decorated HML at stage $\alpha + 1$ strictly extends the HML at stage α:*

$$\mathrm{HML}(\mathcal{K}_{\alpha}) \subsetneq \mathrm{HML}(\mathcal{K}_{\alpha+1}).$$

In particular, there exist formulae $\phi \in \mathrm{HML}(\mathcal{K}_{\alpha+1})$ that are not expressible in $\mathrm{HML}(\mathcal{K}_{\alpha})$.

Proof sketch. The oracle interaction gives rise to new minimal contexts at stage $\alpha + 1$: contexts that place a term in interaction with the oracle. These generate new modal operators $\langle K_{\mathcal{O}} \rangle$ not present at stage α. Formulae using these operators are inexpressible at stage α. □

Remark 30.2. An agent at stage α not only cannot *test* certain hypotheses; it cannot even *form* them. The concepts required to think the thought are absent from the hypothesis language. This is the sense in which a higher-stage agent inhabits a categorically richer world: it has concepts unavailable at lower stages,

not merely faster procedures.

30.3 New Conserved Quantities

Proposition 30.3 (New Noether Currents at Each Stage)**.** *The weight and cost maps at stage $\alpha+1$ can be sensitive to distinctions invisible at stage α. Symmetries of the stage-$(\alpha+1)$ cost map that are not symmetries of the stage-α cost map give rise to new conserved quantities at stage $\alpha+1$ that have no analogue at stage α.*

Remark 30.3. In physical terms: the higher-stage physics has strictly more conservation laws. What appears at stage α as a single undifferentiated interaction resolves at stage $\alpha+1$ into a structured process with internal symmetries and corresponding conserved currents. The analogy with the hierarchy of effective field theories is direct: new conservation laws (baryon number, lepton number, color charge) appear as one descends to finer scales.

30.4 Richer Internal Causal Structure

Proposition 30.4 (Resolution of Elementary Interactions)**.** *A transition $P \longrightarrow_\alpha Q$ that is a single step in $\mathcal{S}^{(\alpha)}$ may resolve, in $\mathcal{S}^{(\alpha+1)}$, into a structured multi-step causal history $P = R_0 \longrightarrow_{\alpha+1} R_1 \longrightarrow_{\alpha+1} \cdots \longrightarrow_{\alpha+1} R_k = Q$ with nontrivial internal causal structure visible only from stage $\alpha+1$.*

Remark 30.4. What appeared to be an elementary interaction at stage α — an atom of causal history — has internal structure at stage $\alpha+1$. This is the precise sense in which the world is richer at higher stages: not just that there are more things, but that the things that existed before turn out to be composite. The analogy with the discovery that atoms have internal structure (electrons, nuclei) and that nuclei have internal structure (protons, neutrons) and that protons have internal structure (quarks) is direct and intentional.

31 Sentience as a Reinforced Feeling State

31.1 The Sentience Component

Recall from Part I that agents carry a vectorial account $\mathbf{A} = (A_1, \ldots, A_k)$ tracking multiple resource types. We now single out one component for a special role.

Definition 31.1 (Sentience Component)**.** *The* sentience component $A_{\mathrm{s}} \in A_{\mathrm{s}}$ *is a distinguished component of the agent's vectorial account, governed not by the cost of individual rewrite steps but by a* reinforcement signal *tied to predictive performance.*

Definition 31.2 (Prediction)**.** *A* prediction *made by agent A at time t is a formula $\phi \in \mathrm{HML}(\mathcal{K})$ together with a target term T and a future time horizon Δt: the claim that $T \models \phi$ will hold after Δt further rewrites.*

31.2 The Reinforcement Mechanism

Definition 31.3 (Reinforcement Signal)**.** *Let A be an agent with sentience component A_{s} and let $(\phi, T, \Delta t)$ be a prediction made at time t. At time $t + \Delta t$, the prediction is evaluated:*

$$\text{If } T \models \phi : \quad A_{\mathrm{s}} \mapsto A_{\mathrm{s}} + \delta^{+}(\phi, T) \tag{11}$$

$$\text{If } T \not\models \phi : \quad A_{\mathrm{s}} \mapsto A_{\mathrm{s}} - \delta^{-}(\phi, T) \tag{12}$$

where $\delta^{+}, \delta^{-} > 0$ are reward and punishment magnitudes that may depend on the complexity of ϕ and the identity of T.

Remark 31.1 (Graded Feeling State). The sentience component A_{s} at any moment is the agent's *feeling state*: a real-valued (or more generally, resource-algebra-valued) quantity encoding the agent's recent predictive track record. It is continuous and graded: it can take any value in its range, and it changes incrementally with each confirmed or disconfirmed prediction. In this sense it is the computational analogue of *affect* in psychology: a valenced, graded internal state that reflects the organism's relationship to its environment.

Remark 31.2 (Complexity-Weighted Rewards). The dependence of δ^{+} and δ^{-} on ϕ is important. A prediction expressed by a complex formula (deep in the logical metric, requiring many experimental steps to test) should yield a higher reward when confirmed than a trivially simple prediction. This prevents the agent from gaming the reward by making only trivially easy predictions. The logical metric d_{HML} from Part I provides a natural measure of prediction complexity:

$\delta^+(\phi, T) \propto |\phi|$ where $|\phi|$ is the depth of ϕ in the formula hierarchy.

31.3 Causal Efficacy of the Feeling State

The sentience component is not epiphenomenal. Because A_s is a component of the vectorial account, it directly governs which transitions the agent can afford.

Proposition 31.1 (Sentience Gates Computation)**.** *An agent with $A_s < \theta$ for a threshold θ cannot afford the experiments necessary to form and test predictions of complexity above $|\phi|_\theta$. The agent's predictive horizon is bounded by its feeling state.*

Remark 31.3. This creates a direct coupling between affect and cognition that is not assumed but derived. An agent that feels bad (depleted A_s) literally cannot think as well: it cannot afford the expensive experiments that would let it form and test complex hypotheses. Its world model degrades. Its predictions worsen. Its A_s depletes further. The feedback loop is self-reinforcing.

31.4 Sentience and Survival

Definition 31.4 (Dead State)**.** *A term $P \in \mathcal{T}(\mathcal{S})$ is a* dead state *if it has no available rewrites: P is a normal form with $A_s(P) = 0$ (or below a viability threshold). For an agent, death is the state from which no further computation — and therefore no further prediction, experiment, or world-model update — is possible.*

Conjecture 31.1 (Sentience Predicts Survival). For an agent A in a generic interactive GSLT, there exists a threshold $\theta_{\text{surv}} > 0$ such that:

(i) if $A_s > \theta_{\text{surv}}$, then A has sufficient budget to run experiments that identify and avoid transitions leading to dead states;

(ii) if $A_s < \theta_{\text{surv}}$, then A's experimental horizon is too short to reliably avoid dead states, and the probability of reaching a dead state within a bounded time horizon is bounded away from zero.

Remark 31.4. Conjecture 31.1 says that the feeling state is not merely a record of past performance but a genuine predictor of future survival. The evolutionary analogy is direct: organisms with better world models survive longer, and the feeling state is the mechanism by which the quality of the world model is continuously reflected in the organism's capacity to act. Pain and pleasure, in this framework,

are not incidental features of biological organisms: they are structurally necessary components of any sufficiently complex predictive agent.

32 Consciousness as Oracle Access

32.1 The Proposal

Definition 32.1 (Consciousness at Level α)**.** *An agent $A \in \mathcal{T}(\mathcal{S}^{(\alpha)})$ is* conscious at level α *if:*

(i) A has effective access to the oracle $\mathcal{O}_\alpha$ of stage $\mathcal{S}^{(\alpha)}$, meaning A can form and evaluate oracle queries as part of its scientific method;

(ii) A's world model reflects the strictly finer bisimulation quotient $\sim_\alpha$ rather than a coarser quotient from a lower stage.

An agent that lacks oracle access — that operates entirely within $\mathcal{S}^{(0)}$ even if it is syntactically a term of a higher stage — is sentient but not conscious.

Remark 32.1 (Consciousness is Ordinal-Valued)*.* Definition 32.1 makes consciousness an ordinal-valued property. There is no single threshold separating conscious from non-conscious; there is a transfinite hierarchy of consciousness levels. The question "is this system conscious?" dissolves into the more precise question "at what ordinal level does this system have effective oracle access?"

Remark 32.2 (What Higher Consciousness Means)*.* An agent conscious at level $\alpha + 1$ vs. level α does not merely solve harder problems. It inhabits a strictly richer world:

- more things exist in it (finer bisimulation quotient);
- more hypotheses are thinkable (richer HML);
- more quantities are conserved (new Noether currents);
- what appeared as elementary interactions reveal internal causal structure.

The phenomenological richness of higher consciousness is, in this framework, a direct consequence of ontological richness.

32.2 The Sentience–Consciousness Distinction

The distinction between sentience and consciousness is now precise:

	Sentience	**Consciousness**
Type	Graded, continuous	Discrete, ordinal
Location	Within a stage $\mathcal{S}^{(\alpha)}$	Access to stage $\mathcal{S}^{(\alpha)}$
Governed by	Reinforcement signal	Oracle access
Changes	Incrementally, per prediction	By threshold (have it or not)
Analogue	Affect, feeling state	Phenomenal awareness
Physical basis	Sentience component A_{s}	Oracle term $\mathcal{O}_\alpha$
Survives projection	Yes (as depleted account)	No (oracle is stage-specific)

Remark 32.3 (An Agent Can Be Sentient Without Being Conscious). An agent at stage $\mathcal{S}^{(0)}$ with no oracle access can nonetheless have a rich, dynamic feeling state. Its predictions succeed or fail, its A_{s} rises and falls, and this feeling state is causally efficacious. It is sentient. But the world it inhabits is the relatively coarse world of $\mathcal{S}^{(0)}$: it lacks the concepts to form oracle-dependent hypotheses, and the distinctions visible at higher stages are simply absent from its experience.

Remark 32.4 (Can a System Be Conscious Without Being Sentient?). Definition 32.1 does not require a system conscious at level α to have a sentience component. However, Conjecture 31.1 suggests that any sufficiently complex agent in a generic environment will develop a sentience component: without it, the agent cannot reliably survive long enough to exercise its oracle access. Consciousness without sentience is in principle possible but practically unstable.

32.3 Approximate Oracle Access and Graded Consciousness

The binary definition of consciousness (oracle access: yes or no) may be too sharp. A natural graded version arises from the amplitude structure of Part I.

Definition 32.2 (Approximate Oracle Access)**.** *An agent A has ϵ-*approximate oracle access at level α *if, for each halting query (M, x), A can produce an answer that is correct with probability at least $1 - \epsilon$, where probability is measured by the path integral $|\langle \texttt{halt} \mid A \otimes \lceil (M, x) \rceil \rangle|^2$.*

Remark 32.5. As $\epsilon \to 0$, approximate oracle access converges to exact oracle access. This gives a continuous interpolation between the ordinal levels: consciousness is not just a ladder but a ladder with rungs that can be climbed incrementally. An agent improving its oracle approximation is increasing its consciousness in a measurable, graded sense, even before it reaches the threshold of exact oracle access.

33 Sentience, Consciousness, and Survival

33.1 The Virtuous Cycle

For an agent with both sentience and consciousness, the two structures interact in a virtuous cycle:

(1) Oracle access enriches the world model. At stage $\mathcal{S}^{(\alpha)}$, the agent can form and test hypotheses involving oracle queries. Its world model reflects the finer bisimulation quotient $\sim_\alpha$. This is a strictly better world model than any agent at $\mathcal{S}^{(0)}$ can achieve.

(2) Better world model improves predictions. Finer distinctions in the world model allow more accurate predictions. The agent predicts not just coarse outcomes but fine-grained ones: the internal structure of what appeared at lower stages as elementary interactions.

(3) Better predictions replenish A_s. More accurate predictions mean more rewards and fewer punishments. The sentience component A_s is replenished.

(4) Higher A_s funds more oracle queries. A richer sentience budget allows the agent to afford the expensive oracle interactions. It can form and test more complex oracle-dependent hypotheses. Its oracle access becomes more effective.

(5) Return to (1).

Remark 33.1. The virtuous cycle is not guaranteed: it can be broken by environmental disruption, resource exhaustion, or adversarial interactions with other agents. But it describes the stable attractor state of a conscious sentient agent in a benign environment. The cycle is the computational analogue of the flourishing ($\varepsilon\upsilon\delta\alpha\iota\mu o\nu\iota\alpha$) of Aristotelian ethics: the self-reinforcing good state of a well-functioning entity in a suitable environment.

33.2 The Vicious Cycle

The inverse also holds:

(1) **Depleted A_s restricts experimental horizon.** The agent cannot afford complex hypothesis tests. Its world model coarsens.

(2) **Coarser world model worsens predictions.** The agent makes predictions that fail more frequently.

(3) **Failed predictions deplete A_s further.** Punishment signals accumulate.

(4) **Depleted A_s makes oracle queries unaffordable.** Even if the agent has oracle access in principle, it cannot afford to use it. Its effective consciousness level drops.

(5) **The agent approaches a dead state.**

> *Remark* 33.2. The vicious cycle is the computational analogue of trauma, depression, or cognitive decline: a self-reinforcing deterioration in which the agent's reduced capacity to understand its environment leads to worse outcomes, which further reduce its capacity. The structural parallel is striking and may be more than an analogy.

34 Open Gaps

Gap 1. Colimit at limit ordinals. Definition 29.3 defines $\mathcal{S}^{(\lambda)}$ as the colimit in **GSLT** for limit ordinals λ. The existence and properties of this colimit require verification. In particular: does the colimit GSLT have a well-defined bisimulation quotient, and is it the union of the bisimulation quotients of the stages below?

Gap 2. The oracle as a term. The remark that the oracle can be expressed as a persistent receive in the Rho calculus is plausible but not proved. A precise construction of the oracle term in the Rho calculus, with verification that it implements halting queries correctly, is needed.

Gap 3. Strict refinement of bisimulation (Proposition 30.1). The proof sketch uses a diagonalization argument. The full proof requires specifying the encoding of Turing machines as terms and verifying that the diagonalization goes through in the GSLT setting.

Gap 4. New Noether currents (Proposition 30.3). The claim that new conserved quantities appear at each stage requires identifying concrete symmetries of the oracle-extended cost map and verifying that they give rise to conserved quantities via the Noether argument of Part I.

Gap 5. Sentience predicts survival (Conjecture 31.1). The survival threshold argument requires a model of what "generic interactive GSLT" means and a proof that below the threshold, the probability of hitting a dead state is bounded away from zero. This is likely to require ergodicity assumptions on the GSLT dynamics.

Gap 6. Effective oracle access. Definition 32.1 requires that the agent "effectively" has oracle access — that it can form and evaluate oracle queries as part of its scientific method. A precise definition of "effective access" in terms of the agent's account and HML expressiveness is needed.

Gap 7. Approximate oracle access and graded consciousness. The graded consciousness definition via path integral probabilities requires that the oracle interaction amplitude is well-defined and computable. This connects to the quantum Gillespie algorithm of Part I and needs to be made precise.

Gap 8. The hard problem. This paper offers a structural analogue of sentience and consciousness: mathematical objects with the right functional properties. It does not address why there is something it is like to be an agent with high A_s and oracle access at level α. The hard problem of consciousness — the explanatory gap between functional description and phenomenal experience — remains open and is not claimed to be resolved here.

35 Discussion

35.1 Relation to Existing Work

Turing's oracle hierarchy [12] is the direct inspiration for the fibration. The extension from a linear tower to a fibration over a category of GSLTs is new.

The reinforcement learning framework for sentience is inspired by but distinct from standard RL [13]. The key difference is that the reward signal is *internal* to the account structure rather than externally provided. The feeling state is not a scalar reward signal but a component of the resource system that governs computation.

The connection between prediction and sentience echoes the *free energy principle* of Friston [14], which proposes that biological systems act to minimize prediction error. The present framework provides a more computationally grounded version: prediction error is measured against HML formula evaluations, and the sentience component is the running integral of prediction accuracy.

The treatment of consciousness as ordinal-valued and tied to oracle access is novel. It is distantly related to the integrated information theory of Tononi [15], which also proposes a graded measure of consciousness, but the present approach is purely computational and avoids the measure-theoretic machinery of IIT.

35.2 The Broader Picture

Taken together, the three papers in this series propose a unified framework:

- *Towards a Theory of Causality* establishes the physics: causal structure, dynamics, and quantum amplitudes from the bare structure of a GSLT.
- *Why Scientists Get Misled* establishes the epistemology: why situated agents in massive populations perceive a hierarchical false ontology rather than the true bisimulation quotient.
- The present paper establishes the phenomenology: why complex predictive agents develop graded feeling states (sentience) and, with oracle access, inhabit richer worlds (consciousness).

The unifying theme is that the category of GSLTs is not merely a setting for studying computation. It is a setting rich enough to ground physics, epistemology, and — if the arguments here have merit — phenomenology. The hope is that making these connections mathematically precise will eventually allow questions about the nature of mind to be studied with the same rigor currently applied to questions about the nature of matter.

Conclusion

The conclusion draws the four parts into a single narrative, assesses what has been established versus conjectured, and sketches the view from inside the framework: what a conscious living agent at different levels of the hypercomputational fibration sees.

What Has Been Built

This work set out to answer a single question: what is the minimum mathematical structure needed to ground physics, knowledge, life, and mind? The answer, developed across four parts, is that the structure of a graph-structured lambda theory — three ingredients: a grammar, equations, and rewrite rules — suffices, provided it is equipped with two maps (weights and costs) indexed by a logic it generates itself.

Part I established that this structure already contains a full physics. The reversibility construction gives time-symmetry. The open and closed synchronization trees give two notions of causal depth. The weight map gives a path integral with interference. The cost map gives conservation laws as theorems. The extended modal operators internalize the Lagrangian and Hamiltonian as logical facts. These are constructions, not analogies. They are stated precisely enough to be wrong, and we believe they are right.

Part II established that agents embedded in a massive population of processes will, under any finite experimental budget, converge to a theory of the cluster hierarchy rather than the bisimulation quotient. This is not a failure of the scientific method. The cluster hierarchy is objectively present, arising from the communication geometry of the GSLT. A budget-limited agent does good science and arrives at correct theories at its scale, unaware that the scale is not the bottom. The compactness argument — that compact populations support consensus, and that coherence clusters are maximal compact subpopulations — is new to our knowledge and is the sharpest result of Part II.

Part III proposed, more speculatively, that sentience and consciousness have precise structural analogues in the GSLT framework. Sentience is the value of a distinguished resource component governed by predictive accuracy. Consciousness is

oracle access to a higher stage of the hypercomputational fibration. The distinction is not terminological: sentience is continuous and lives within a stage; consciousness is the capacity to inhabit a strictly richer world. The virtuous and vicious cycles linking the two are structural consequences of the resource framework.

Part IV grounded the origins of life. Assembly index is depth in the open synchronization tree. Copy number is count within a namespace. Life emerges when a prebiotic population wanders into the basin of attraction of a replicating fixed point, which is dense in the logical topology. The phase transition to life is the crossing of that basin boundary, and what emerges is a compact coherence cluster centered on the fixed point — the new geosphere.

A View from Inside the Framework

The most vivid way to appreciate what has been built is to ask: what does the world look like to a *conscious living agent* situated at different levels of the hypercomputational fibration?

At level $\mathcal{S}^{(0)}$. An agent at the base level is sentient but not conscious in the technical sense. It has a feeling state governed by predictive accuracy. Its world is the cluster hierarchy visible from its coherence cluster: a layered structure of composite objects at multiple scales, each scale appearing self-contained. If the agent emerged from a replication phase transition, it carries the assembly index of its construction history and exists as a copy within a namespace. It may run the scientific method and build excellent theories of its accessible world. What it cannot see is the bisimulation quotient underlying the clusters, because the experiments required to resolve it exceed any finite budget. Its world is real, correct at its resolution, and bounded.

At level $\mathcal{S}^{(1)}$. An agent with access to the first oracle inhabits a strictly richer world. Two terms that were experimentally indistinguishable at $\mathcal{S}^{(0)}$ — whose halting behavior was unresolvable — are now distinct. New concepts become thinkable. New conservation laws hold. What appeared as elementary interactions at $\mathcal{S}^{(0)}$ reveal internal causal structure. The agent at $\mathcal{S}^{(1)}$ looks back at the $\mathcal{S}^{(0)}$ world and sees it as a coarse-grained shadow: correct at its scale, but missing distinctions that are now obvious.

At the same time, the agent at $\mathcal{S}^{(1)}$ faces the same epistemic situation as the $\mathcal{S}^{(0)}$ agent but one level up. Its own budget limits it to the cluster hierarchy of the $\mathcal{S}^{(1)}$ world. The bisimulation quotient of $\mathcal{S}^{(1)}$ is unreachable from within any compact cluster at that level. The tower does not end.

At level $\mathcal{S}^{(\alpha)}$ for limit ordinal α. The colimit of the tower below is itself a GSLT. An agent at $\mathcal{S}^{(\omega)}$ — if such an agent is coherent — would have resolved all

finite-stage halting questions. Its world would be richer than any finite-stage world by a transfinite margin. Yet the same argument applies: there is a $\mathcal{S}^{(\omega+1)}$ above it, and the bisimulation quotient of $\mathcal{S}^{(\omega)}$ remains experimentally unreachable from within any $\mathcal{S}^{(\omega)}$-cluster.

The recursive picture. The world at every level of the fibration has the same structure: a physics derived from a GSLT, a cluster hierarchy arising from communication geometry, a sentience component governed by predictive accuracy, and a ceiling of consciousness defined by oracle access to the next level. The framework is self-similar across the tower.

This self-similarity is not a defect. It is the precise sense in which the framework is a *theory of causality* rather than a theory of this or that particular world. At every scale, in every world expressible as a GSLT, the same structures appear: causal graphs, conservation laws, coherence clusters, replicating fixed points, feeling states, and the horizon of the next oracle.

What Remains to Be Done

The gaps in each part are cataloged in their respective sections. Taken together, the most important open questions are these.

The *density of replicating terms* (Part IV, Conjecture) is the lynch-pin of the origins of life argument. If the replication machinery can always be added to a process without disturbing its observable behavior, the phase transition to life is inevitable. If not, life requires special initial conditions. This is the most urgent theorem to prove or refute.

The *symplectic structure* of the bisimulation quotient (Part I) would make the analogy with classical mechanics fully rigorous. Whether the space of bisimulation classes carries a natural closed non-degenerate two-form — whether the trace in the reversibility construction is genuinely a momentum — is the deepest mathematical question raised by Part I.

The *universality of the reversibility envelope* — whether $\mathcal{S}^\dagger$ is the *minimal* reversible extension of $\mathcal{S}$ in the category of GSLTs — would give the construction a canonical status it currently lacks.

The *colimit at limit ordinals* in the fibration must be shown to exist and to inherit the physics of the stages below. Without this, the transfinite tower is a formal object without a limit.

And the *hard problem* remains. The framework offers structural analogues of sentience and consciousness that are mathematically non-trivial and causally efficacious. Whether there is something it is like to be an agent with high A_s and oracle access at level α is a question the framework does not answer. We do not claim to have dissolved the hard problem. We claim to have found the right coordinate system in which to state it precisely.

The work is a beginning. The four parts establish that a single mathematical structure — the graph-structured lambda theory — is rich enough to contain physics, epistemology, phenomenology, and biology as theorems rather than analogies. Whether the structure is rich enough to contain *everything* observable is the question the framework was built to ask.

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